

# Rounding switching controls under a hard switch budget: sharp minimax bounds and exact algorithms

## Abstract

We study cumulative-error rounding of measurable simplex-valued controls when the integer control has a hard bound on its number of switches. We determine the continuous minimax error through three switches whenever the number of modes exceeds the number of activation blocks. A heavy-mode reduction and sharp one-sided reach bounds give matching extremizers; the four-block proof uses exact certificates covering every mode count. Mode removal yields general-budget bounds, enlarged exact plateau ranges, and the sharp first mode-count correction to the dimension-free limit. On finite grids we derive an explicit arbitrary-grid one-switch minimax formula and exact three-mode values from a complete floor-history representation. These include the continuous boundary value with two switches and counterexamples to specific published finite-grid claims. For an input with  $n$  modes on  $N$  grid cells, we give an exact one-switch algorithm using  $O(nN)$  rational operations and a fixed-budget subset algorithm with linear arithmetic dependence on the number of modes. A sharp switch-preserving grid transfer provides additive coarse-grid certificates with quantified input perturbation. Exact rational checks and reproducible public-profile computations accompany the proofs.

## 1 Introduction

A relaxed control can divide activity among several operating modes, whereas an implementable switching control activates one mode at a time. Combinatorial integral approximation measures the resulting discrepancy by the largest absolute difference between cumulative mode allocations. A fine grid permits small discrepancy when switches are unrestricted. With a hard switch budget, refining the grid only improves the placement of finitely many changes; it cannot remove the approximation error caused by the budget itself. The relevant questions are therefore quantitative: what error is unavoidable, which relaxed inputs attain it, and how can one compute an optimal or certifiably near-optimal schedule?

We answer these questions for measurable simplex-valued controls on a horizon  $[0, T]$ , with  $n$  modes and at most  $s$  switches. The initial mode is free, repeated modes are allowed, and the default model has no dwell or transition restrictions. For an input cumulative allocation  $A$ , write  $\text{OPT}_s(A)$  for the smallest cumulative discrepancy among continuous-time schedules and  $F_{n,s}(T) = \max_A \text{OPT}_s(A)$  for its worst-case value. A superscript  $\mathcal{T}$  restricts switching times to a specified grid. The distinction between an input optimum and the maximum over inputs is essential: a sharp bound on the discretization error of each input does not establish a sharp gap between two worst-case values.

### 1.1 Results and proof strategy

Our main continuous minimax formula is

$$F_{n,s}(T) = T \max \left\{ \frac{1}{s+2}, \frac{1}{n((n/(n-1))^{s+1} - 1)} \right\}, \quad 0 \leq s \leq 3, \quad n \geq s+2. \quad (1)$$

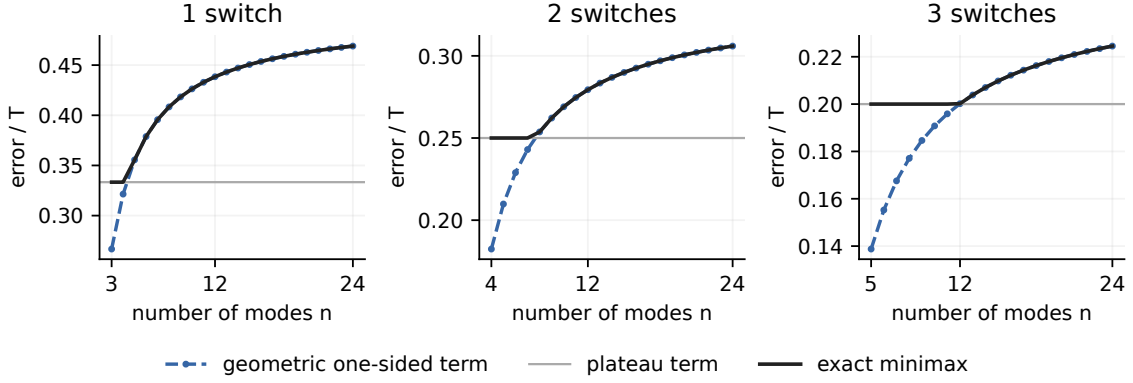


Figure 1: The exact continuous minimax value divided by  $T$ , its plateau term, and the geometric one-sided uniform term for one, two, and three switches, on their stated mode ranges. Lines connect integer mode counts.

A uniform input supported on  $s + 2$  modes attains the first term. The second is the sharp one-sided error for the uniform input on all  $n$  modes. This geometric term exceeds the plateau term starting at  $n = 5, 8, 12$  for  $s = 1, 2, 3$ , respectively. Figure 1 illustrates these transitions. The formula includes all measurable inputs, and its lower proof allows competing schedules to repeat modes.

The main reduction separates a heavy-mode case from a one-sided discrepancy problem. If some terminal mode mass exceeds  $E$  and  $T \leq (s + 2)E$ , an integral-prefix rounding and a reordering of the first repetition produce an  $s$ -switch schedule of full error at most  $E$ . Otherwise, a one-sided bound suffices. We prove sharp reach bounds using distinct modes through four blocks. The two- and three-block arguments are analytic. The four-block argument uses an explicitly defined necessary-condition linear program, exhaustive symmetry reduction, and exact rational and polynomial dual certificates covering every  $n \geq 5$ . Floating-point optimization is not a proof dependency. At the boundary  $n = k = 3$ , a constructed input has an exact three-block one-sided optimum strictly worse than the uniform input, even when repeated modes are allowed. Thus the spare-mode hypothesis in this reach method has mathematical content.

For arbitrary block count  $k = s + 1 < n$ , removing either a largest-mass or a smallest-mass mode propagates existing bounds. The resulting closed formula and its four-block seeded improvement yield exact plateau ranges and

$$F_{n,k-1}(T) = \frac{T}{k} - \frac{(k+1)T}{2kn} + O_k(Tn^{-2}) \quad (n \rightarrow \infty).$$

The leading dimension-free bound follows from classical integral-prefix rounding; the first mode-count correction is sharp. Equal-terminal-mass inputs have additional guarantees, including an exact instance value in a band next to  $n = k$ . The remaining general five-block reach question is isolated in Appendix A: an earlier event relaxation admits a nonphysical counterexample, whereas a fully ordered chamber has a matching exact certificate. Neither observation is used as a proof for other chambers.

On finite grids, a separate extremal analysis gives an explicit one-switch minimax formula on every rational grid for  $n \geq 3$ , evaluable in  $O(N^2)$  rational operations, with a worst input having two component types and at most three time phases. Three-mode unit grids admit a finite floor-history representation valid for every length. It gives exact values on five, six, and seven cells, including  $F_{3,3}^{\text{unit}}(7) = 4/3$ . Its five-cell consequence closes the continuous

Table 1: Main scopes. Here  $k = s + 1$ , and a unit grid has horizon  $N$ .

| Problem                                                   | Result                                                 | Reference    |
|-----------------------------------------------------------|--------------------------------------------------------|--------------|
| Continuous minimax, $0 \leq s \leq 3$ ,<br>$n \geq s + 2$ | Exact formula (1)                                      | Theorem 7.2  |
| General budgets, $n > k$                                  | Closed and seeded upper bounds; sharp first correction | Section 9    |
| Three-mode boundaries                                     | $F_{3,2} = T/5$ ; $T/7 \leq F_{3,3} \leq T/6$          | Section 12   |
| One-switch grid minimax, $n \geq 3$                       | Explicit formula on arbitrary grids                    | Theorem 11.1 |
| Three-mode unit grids                                     | $(N, s) = (5, 2), (6, 3), (7, 3)$ : values 1, 1, 4/3   | Section 12   |
| Supplied grid input                                       | Exact one-switch and fixed-budget algorithms           | Section 13   |
| Continuous-to-grid transfer                               | Sharp uniform coefficient; additive certificates       | Section 14   |

boundary value  $F_{3,2}(T) = T/5$ . For three modes and three switches, the proved continuous range is  $T/7 \leq F_{3,3}(T) \leq T/6$ ; an exact value is not asserted. These results both sharpen and correct specific finite-grid claims in Sager and Zeile [9]. The source comparisons distinguish a false equality, an upper bound that fails, and a lower bound that fails.

The algorithmic results solve supplied inputs. One-switch optimization takes  $O(nN)$  arithmetic operations on an arbitrary grid. For a fixed number  $k$  of nominal blocks, a subset-partition dynamic program assigns blocks to modes, allowing repeats and checking specified minimum dwell times on maximal runs. Enumerating the block boundaries gives polynomial time for every fixed switch budget, with linear dependence on  $n$  in the arithmetic bound. A supported chronological grid transfer preserves the switch budget and changes cumulative occupations by strictly less than one uniform cell width  $h$ . Its coefficient one is sharp uniformly over modes and budgets. Combining this transfer with exact coarse optimization yields a computable additive certificate for the continuous optimum, with explicit input-quantization bounds. Dwell restrictions require separate treatment: an elementary example has a fixed positive grid gap on arbitrarily fine odd grids.

## 1.2 Relation to earlier work

The CIA decomposition and switch-constrained approximation subproblem were introduced by Sager, Jung, and Kirches [10], including a mixed integer linear formulation and a tailored branch-and-bound algorithm. Sager and Zeile [9] study a total-variation constraint on the integer control and derive worst-case bounds and maximum dwell rounding. Their multi-mode conjecture is the closest direct comparison for the minimax results here. The continuous lower constructions and the precise published finite-grid statements are credited where they are used or corrected; see Sections 3 and 12.

Integral flow and matching are established tools for rounding partial sums; Knuth’s two-way rounding theorem [7] is a classical example. Besthorn and Kirches [2, Corollary 2.7] prove support-preserving CIA rounding with discrepancy at most one uniform grid width. Our transfer proof gives this construction explicitly and adds the chronological argument that preserves a supplied schedule’s switch count, the strict bound, and a sharp instance-wise discretization family. The half-grid motivation preceding the conjecture in [9, p. 615] and [13, p. 139] is already a proposed discretization comparison; it is not a proved universal theorem. Section 14 identifies its valid special cases and an explicit counterexample to its unrestricted instance-wise interpretation.

Exact graph algorithms for switching costs, minimum dwell times, and support restrictions already exist. In particular, the shortest-path formulation of Besthorn, Hansknecht,

Kirches, and Manns [3] gives switching-cost-aware rounding (SCARP), with a complexity bound exponential in the mode count. Zeile, Robuschi, and Sager [11] analyze minimum dwell rounding, and switching-time enumeration is discussed in [13, Section 6.4.3]. More recently, Abbasi-Esfeden, Plate, Sager, and Swevers [1] propose a flexible dynamic-programming-inspired CIA method with dwell constraints; their introduction explicitly states that it does not generally guarantee global optimality because optimal substructure is lost. The distinction in the present exact algorithms is the CIA-specific dominance and subset reduction, with its proved dependence on the mode count for a fixed small budget. Dynamic programming and dwell-constrained rounding themselves are established.

Several approximation results concern grids whose refinement permits the number of switches to grow. Tight sum-up-rounding bounds [5], compactness and convergence of CIA decompositions [6], and adaptive nonuniform refinement for SCARP [4] provide this broader setting. Our coarsening certificate concerns the best approximation with a fixed switch budget and a prescribed additive accuracy, including exact integration across nonaligned input and output grids. It does not assert convergence to the relaxed control at fixed budget. Finally, cumulative discrepancy can bound state error under appropriate regularity assumptions, as in [12, Theorem 1 and Corollary 1]. Here the proved and measured quantities are discrepancy, switching feasibility, and optimization cost. Claims about nonlinear state constraints, economic objectives, or closed-loop stability require the corresponding dynamics and assumptions.

### 1.3 Organization

Section 2 gives the model and attainment and stability properties. Sections 3–7 establish uniform-input formulas and the sharp continuous results. The structural counterexamples in Section 8 delimit the constructions. Sections 9 and 10 develop general budgets and retain useful input-dependent bounds. The finite-grid minimax results follow in Sections 11 and 12. Sections 13 and 14 give the instance algorithms and coarsening certificates. Section 15 reports reproducible exact computations. Section 16 identifies the remaining research questions, with the detailed higher-reach analysis in Appendix A.

## 2 Problem and basic properties

A relaxed switching control allocates one unit of activity among finitely many modes. Rounding selects one mode at a time. The cumulative allocation error is the objective of combinatorial integral approximation (CIA), the rounding subproblem in a decomposition of mixed-integer optimal control. A restriction on the number of mode changes is a hard constraint on that subproblem. Sager and Zeile [9] study this formulation, develop maximum dwell rounding, and give sharp bounds and a conjecture for its worst-case error. Our focus is the extremal rounding problem itself. In particular, a bound on cumulative allocation error is not an assertion that the same schedule optimizes an underlying nonlinear control objective.

### 2.1 Controls, schedules, and the two error criteria

Fix a horizon  $T > 0$  and an integer number of modes  $n \geq 2$ . Throughout, switch budgets  $s$  are nonnegative integers, activation-block counts  $k$  are positive integers, and grid sizes  $N$  are positive integers. Write  $[n] = \{1, \dots, n\}$  and

$$\Delta_n = \left\{ a \in \mathbb{R}^n : a_i \geq 0, \sum_{i=1}^n a_i = 1 \right\}.$$

A relaxed control is a Lebesgue-measurable map  $\alpha : [0, T] \rightarrow \Delta_n$ , considered up to equality almost everywhere. Its cumulative allocation and terminal masses are

$$A_i(t) = \int_0^t \alpha_i(u) du, \quad m_i = A_i(T). \quad (2)$$

Thus  $A_i$  is nondecreasing and 1-Lipschitz,  $A_i(0) = 0$ , and  $\sum_i A_i(t) = t$ . Conversely, every vector of functions with these properties is the cumulative allocation of such a control: Lipschitz functions are absolutely continuous and differentiable almost everywhere, their derivatives are nonnegative, and differentiation of the sum gives a simplex-valued vector almost everywhere. Denote this class of cumulative allocations by  $\mathcal{A}_n(T)$ . We use either  $\alpha$  or  $A$  to specify an input.

An integer control, or schedule, selects a coordinate vector  $e_i$  on each of finitely many intervals. Its initial activation is free: only changes between two successive active modes count as switches. Its values at isolated times are immaterial. Let  $\mathcal{W}_{n,k}(T)$  be the set of cumulative occupations

$$W_i(t) = \int_0^t \omega_i(u) du$$

of schedules with at most  $k$  positive-length activation blocks, equivalently at most  $s = k - 1$  switches. Unless explicitly stated otherwise, repeated modes are allowed, the initial and final modes are unrestricted, and there are no dwell-time or transition restrictions. A parameterization of this entire set uses a word  $p \in [n]^k$  and ordered times

$$0 = t_0 \leq t_1 \leq \dots \leq t_k = T, \quad \omega(t) = e_{p_j} \quad (t_{j-1} < t < t_j). \quad (3)$$

Zero-length intervals can be deleted and consecutive equal modes can be merged. These operations cannot increase the switch count. Consequently this parameterization includes shorter schedules without imposing distinctness on competing modes.

The full error and the one-sided error are respectively

$$D(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} |A_i(t) - W_i(t)|, \quad D^-(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} (W_i(t) - A_i(t)). \quad (4)$$

The minus superscript refers to the negative part of the discrepancy  $A - W$ ; the displayed one-sided maximum is nonnegative because it includes  $t = 0$ . For a fixed input, define the continuous instance optima

$$\text{OPT}_s(A) = \min_{W \in \mathcal{W}_{n,s+1}(T)} D(A, W), \quad \text{OPT}_k^-(A) = \min_{W \in \mathcal{W}_{n,k}(T)} D^-(A, W). \quad (5)$$

The corresponding minimax values are

$$F_{n,s}(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_s(A), \quad G_{n,k}^-(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_k^-(A). \quad (6)$$

The indexing is intentional:  $F$  counts switches, whereas  $G^-$  counts activation blocks, with  $k = s + 1$ . Both infima in (5) and both suprema in (6) are attained, as shown below.

## 2.2 Fixed grids and endpoint evaluation

A grid is a strictly increasing sequence  $\mathcal{T} = (x_0, \dots, x_N)$  with  $x_0 = 0$ ,  $x_N = T$ . Its cell lengths are  $\Delta_j = x_j - x_{j-1}$  and  $\Delta = \max_j \Delta_j$ . Let  $\mathcal{A}_n(\mathcal{T})$  consist of cumulative allocations of controls constant on each cell, and let  $\mathcal{W}_{n,k}(\mathcal{T})$  restrict the switching times to grid points. Set

$$\text{OPT}_s^{\mathcal{T}}(A) = \min_{W \in \mathcal{W}_{n,s+1}(\mathcal{T})} D(A, W), \quad F_{n,s}^{\mathcal{T}} = \max_{A \in \mathcal{A}_n(\mathcal{T})} \text{OPT}_s^{\mathcal{T}}(A). \quad (7)$$

The instance optimum is defined even for  $A \in \mathcal{A}_n(T)$  that is not piecewise affine. The grid minimax in (7) does restrict the input to cellwise constant controls. Analogously, use  $\text{OPT}_k^{-,\mathcal{T}}$  and  $G_{n,k}^{-,\mathcal{T}}$  for the one-sided criterion. On the unit grid  $(0, 1, \dots, N)$  we abbreviate the full minimax by  $F_{n,s}^{\text{unit}}(N)$ . Every grid with cell length  $\Delta$  is obtained from a unit grid by scaling all errors by  $\Delta$ .

For the same input,  $\text{OPT}_s(A) \leq \text{OPT}_s^{\mathcal{T}}(A)$ . This fact alone does not compare the two full minimax values, whose input classes differ. The following observation permits exact evaluation even for merely measurable inputs and will be used throughout.

**Lemma 2.1** (Endpoint monotonicity). *On any interval on which the selected mode is  $p$ , the discrepancy  $A_p - W_p$  is nonincreasing and every other discrepancy  $A_i - W_i$  is nondecreasing. Hence both errors in (4) are determined by the activation-block endpoints. In particular, for a grid schedule they are determined by grid endpoints, even when  $\alpha$  varies inside grid cells.*

*Proof.* The absolutely continuous discrepancies have derivatives  $\alpha_p - 1 \leq 0$  and  $\alpha_i \geq 0$  for  $i \neq p$ , almost everywhere on the interval. Integrating these inequalities proves monotonicity. A monotone real function takes its largest absolute value and each of its one-sided extrema at an endpoint.  $\square$

Replacing a measurable control by its cell averages preserves every grid endpoint  $A_i(x_j)$ . Lemma 2.1 therefore shows that this replacement preserves the error of every grid schedule, and hence its grid instance optimum. This observation does not say that averaging preserves its continuous instance optimum. It does, however, give the minimax comparison

$$F_{n,s}(T) \leq F_{n,s}^{\mathcal{T}}, \quad G_{n,k}^-(T) \leq G_{n,k}^{-,\mathcal{T}}. \quad (8)$$

Indeed, for any measurable input  $A$ , its cell average  $\bar{A}$  satisfies  $\text{OPT}_s(A) \leq \text{OPT}_s^{\mathcal{T}}(A) = \text{OPT}_s^{\mathcal{T}}(\bar{A}) \leq F_{n,s}^{\mathcal{T}}$ . Taking the supremum proves the first inequality; the one-sided argument is identical.

### 2.3 Existence, continuity, and scaling

Equip vector-valued cumulative functions with  $\|A\|_{\infty} = \max_i \max_{t \in [0,T]} |A_i(t)|$ .

**Proposition 2.2** (Attainment and stability). *The sets  $\mathcal{A}_n(T)$  and  $\mathcal{W}_{n,k}(T)$  are compact in the uniform norm. Each of the four instance optimum maps defined above is 1-Lipschitz in  $A$ . All continuous and grid instance optima and minimax values are attained.*

*Proof.* The cumulative allocations are uniformly bounded and equicontinuous. Their uniform limits preserve monotonicity, the Lipschitz constants, initial values, and the sum identity, and therefore belong to  $\mathcal{A}_n(T)$  by the converse following (2). The Arzelà–Ascoli theorem gives compactness.

For a fixed word  $p$ , the ordered time simplex in (3) is compact. Its cumulative occupations are continuous in those times. Indeed, the useful representation

$$W_i(t) = \mathbf{1}_{\{p_k=i\}}t + \sum_{j=1}^{k-1} (\mathbf{1}_{\{p_j=i\}} - \mathbf{1}_{\{p_{j+1}=i\}}) \min\{t, t_j\}$$

gives  $\left\| W(\cdot; t_1, \dots, t_{k-1}) - W(\cdot; t'_1, \dots, t'_{k-1}) \right\|_{\infty} \leq \sum_{j=1}^{k-1} |t_j - t'_j|$ . Taking the finite union over words proves compactness of  $\mathcal{W}_{n,k}(T)$ . Both error criteria are continuous, so their minima are attained. For either criterion  $d$ , the inequality  $|d(A, W) - d(B, W)| \leq \|A - B\|_{\infty}$  holds for every  $W$ . Taking minima shows the stated Lipschitz property. Compactness of  $\mathcal{A}_n(T)$  then gives attainment of the outer maxima. For grids, there are finitely many schedules and the input is a product of  $N$  compact simplexes, proving the analogous claims.  $\square$

Changing time by  $t = Tu$  is a bijection of both input and schedule classes and multiplies errors by  $T$ . Thus

$$F_{n,s}(T) = TF_{n,s}(1), \quad G_{n,k}^-(T) = TG_{n,k}^-(1). \quad (9)$$

The same statement applies to instance optima and a simultaneously rescaled grid. At  $T = 0$  all errors are zero; results stated for  $T > 0$  extend with this convention. Enlarging the switch budget cannot increase any instance optimum or minimax value.

**Proposition 2.3** (No-switch baseline). *For every  $n \geq 2$ ,  $F_{n,0}(T) = T(1 - 1/n)$ . On any grid,  $F_{n,0}^T = T(1 - 1/n)$  as well.*

*Proof.* Activating mode  $p$  throughout gives error  $\max\{T - m_p, \max_{i \neq p} m_i\} = T - m_p$ , since  $\sum_{i \neq p} m_i = T - m_p$ . Choosing a largest terminal mass gives error at most  $T(1 - 1/n)$ . The uniform control  $\alpha_i = 1/n$  gives equality and belongs to every grid input class.  $\square$

### 3 Uniform inputs and the exact one-switch minimax

We first solve a benchmark family for every budget with fewer activation blocks than modes. This family gives a dimension-dependent obstruction to small error. We then prove that its obstruction, together with an omitted-mode obstruction, determines the full continuous one-switch minimax for at least three modes.

#### 3.1 Exact uniform-input error in continuous time

Write  $A_i^{\text{unif}}(t) = t/n$  and let  $E_{n,s}(T) = \text{OPT}_s(A^{\text{unif}})$ .

**Theorem 3.1** (Uniform controls). *Let  $n \geq 2$  and  $0 \leq s \leq n - 2$ . With  $k = s + 1$  and  $r = n/(n - 1)$ ,*

$$E_{n,s}(T) = T \max \left\{ \frac{1}{n}, \frac{1}{n(r^k - 1)} \right\}. \quad (10)$$

*A schedule using  $k$  distinct modes attains this value, although competing schedules may repeat modes.*

*Proof.* Consider any schedule with  $m \leq k < n$  positive-length blocks and error  $E$ . At least one mode is unused, so  $E \geq T/n$ . Let  $0 = t_0 < \dots < t_m = T$  be its block endpoints. At the end of block  $j$ , the occupation of its active mode  $p_j$  is at least  $t_j - t_{j-1}$ , even if  $p_j$  occurred earlier. The inequality  $W_{p_j}(t_j) - t_j/n \leq E$  implies

$$t_j \leq rt_{j-1} + rE. \quad (11)$$

Iterating from zero yields  $T \leq nE(r^m - 1) \leq nE(r^k - 1)$  and proves both lower bounds.

For attainment, set

$$E_0 = \frac{T}{n(r^k - 1)}, \quad t_j = T \frac{r^j - 1}{r^k - 1} \quad (0 \leq j \leq k),$$

and choose a distinct mode on each block. Before its activation, the selected mode has discrepancy  $t/n$ ; during activation its discrepancy decreases; after activation it increases again. At its block endpoint,

$$A_{p_j}(t_j) - W_{p_j}(t_j) = \frac{t_j}{n} - (t_j - t_{j-1}) = -E_0.$$

Every positive discrepancy is at most  $T/n$ , and no selected mode's negative discrepancy exceeds  $E_0$ . Unused modes also have error at most  $T/n$ . This proves attainment of (10).  $\square$

For one switch, this gives

$$E_{n,1}(T) = T \max \left\{ \frac{1}{n}, \frac{(n-1)^2}{n(2n-1)} \right\} \quad (n \geq 3). \quad (12)$$

For every fixed  $s \geq 0$ , the uniform value tends to  $T/(s+1)$  as  $n \rightarrow \infty$ . Indeed, with fixed  $k = s+1$ ,  $(1 + 1/(n-1))^k - 1 = k/(n-1) + O_k(n^{-2})$ , so  $n(r^k - 1) \rightarrow k$ . This is an obstruction supplied by one input family; it does not presume that this family maximizes the error over all inputs.

### 3.2 A scalar recurrence on uniform grids

Let  $E_{n,s}^{\text{unit}}(N)$  be the grid instance optimum for the uniform control on  $N$  unit cells.

**Theorem 3.2** (Exact integer recurrence). *Under  $n \geq 2$ ,  $0 \leq s \leq n-2$ , and  $N \geq 1$ , define, for integers  $K \geq N$ ,*

$$b_0(K) = 0, \quad b_{j+1}(K) = \left\lfloor \frac{nb_j(K) + K}{n-1} \right\rfloor. \quad (13)$$

*If  $K_*$  is the least such integer for which  $b_{s+1}(K_*) \geq N$ , then  $E_{n,s}^{\text{unit}}(N) = K_*/n$ . Moreover,*

$$E_{n,s}(N) \leq E_{n,s}^{\text{unit}}(N) < E_{n,s}(N) + \frac{n-1}{n}. \quad (14)$$

*Proof.* At integer times, every discrepancy is a multiple of  $1/n$ ; by Lemma 2.1 an optimal error is therefore  $K/n$  for an integer  $K$ . Every admissible schedule omits a mode, giving  $K \geq N$ . The block-end argument (11), with integer block endpoints, gives  $t_j \leq \lfloor (nt_{j-1} + K)/(n-1) \rfloor$ . Monotonicity of this recurrence implies  $t_j \leq b_j(K)$  and hence necessity of  $b_{s+1}(K) \geq N$ . Fewer blocks can be padded by zero-length blocks; alternatively the nondecreasing iterates give the same conclusion.

Conversely, suppose the condition holds. Put  $t_j = \min\{N, b_j(K)\}$  and stop when  $N$  is first reached. Since the recurrence cannot leave zero if  $b_1(K) = 0$ , the assumed condition implies  $b_1(K) \geq 1$ , or  $K \geq n-1$ . It follows that the iterates strictly increase before truncation. Assign a distinct mode to each positive-length block. Its block-end inequality gives  $W_i(t_j) - t_j/n \leq K/n$ . Before and after that block, its discrepancy is nondecreasing, so this bounds its negative part throughout. Positive errors of selected and unused modes are at most  $N/n \leq K/n$ . There are at most  $s+1$  blocks, proving sufficiency.

The predicate is monotone in  $K$ , and  $K = N(n-1)$  already reaches  $N$  in one step. Thus  $K_*$  exists and binary search may use  $N \leq K \leq N(n-1)$ , with  $s+1$  recurrence evaluations per trial. To prove the strict gap, write  $E = E_{n,s}(N)$  and  $C = \lceil nE \rceil$ , and test  $K = C + n - 2 \geq N$ . For every integer  $b$ ,

$$\left\lfloor \frac{nb + C + n - 2}{n-1} \right\rfloor = \left\lceil \frac{nb + C}{n-1} \right\rceil \geq rb + rE.$$

The discrete recurrence therefore dominates the continuous recurrence starting at zero, which reaches at least  $N$  in  $s+1$  steps by (10). Consequently  $E_{n,s}^{\text{unit}}(N) \leq K/n < E + (n-1)/n$ . The lower inequality follows by restricting continuous switching times to integer times for the same input.  $\square$

### 3.3 The continuous one-switch minimax

For two distinct modes  $p, q$ , let  $W^{p,q,\tau}$  activate  $p$  before time  $\tau$  and  $q$  afterward, allowing  $\tau = 0$  or  $T$ . These endpoints include constant schedules.

**Lemma 3.3** (The three-term formula). *For  $n \geq 2$  and  $0 \leq \tau \leq T$ , taking the maximum over omitted terminal masses as zero when  $n = 2$ ,*

$$D(A, W^{p,q,\tau}) = \max \left\{ \max_{i \notin \{p,q\}} m_i, \quad \tau - A_p(\tau), \quad T - m_q - \tau \right\}. \quad (15)$$

*Proof.* The omitted modes increase to their terminal masses. The discrepancy of  $p$  decreases from zero to  $A_p(\tau) - \tau$  and then increases to  $m_p - \tau$ . Its possible positive final error is bounded by  $T - m_q - \tau$ , because  $m_p + m_q \leq T$ . The discrepancy of  $q$  increases to  $A_q(\tau)$  and then decreases to  $m_q - (T - \tau)$ . Its positive error at activation is at most  $\tau - A_p(\tau)$ , because  $A_p(\tau) + A_q(\tau) \leq \tau$ . These comparisons bound every endpoint extremum by the right-hand side. Conversely, the first two displayed terms are actual errors. If the last term is positive it is an actual negative terminal discrepancy; if it is nonpositive it is dominated by the nonnegative second term. This proves the equality, including  $\tau = 0, T$ .  $\square$

**Theorem 3.4** (Exact continuous one-switch minimax). *For every  $n \geq 3$ ,*

$$F_{n,1}(T) = H_n(T) := T \max \left\{ \frac{1}{3}, \frac{(n-1)^2}{n(2n-1)} \right\}. \quad (16)$$

*For  $n = 3, 4$ , three consecutive distinct pure modes, each of duration  $T/3$ , attain the maximum. For  $n \geq 5$ , the uniform control attains it. A schedule with the asserted error bound can be constructed using terminal masses and cumulative allocations at at most two additional times.*

*Proof.* By scaling take  $T = 1$  and put  $E = H_n(1)$ . Directly from the formula,  $1/3 \leq E < 1/2$ .

First suppose some  $m_q \geq E$ . At most one other mode has mass strictly larger than  $E$ : two such modes together with  $q$  would have total exceeding  $3E \geq 1$ . Choose  $p \neq q$  to include that mode if it exists, and choose any other mode otherwise. All omitted masses are at most  $E$ . With  $\tau = \max\{0, 1 - m_q - E\}$  we have  $0 \leq \tau \leq \max\{0, 1 - 2E\} \leq E$ . Thus  $\tau - A_p(\tau) \leq E$  and  $1 - m_q - \tau \leq E$ , and Lemma 3.3 gives a feasible schedule.

It remains to treat the case  $m_i < E$  for every mode. Let  $q$  index a largest mass and  $r$  a second largest, breaking ties arbitrarily. Define

$$t_q = 1 - m_q - E, \quad t_r = 1 - m_r - E, \quad h_i = 1 - m_i - 2E.$$

Since  $m_i < E < 1/2$ , the times satisfy  $0 < t_q \leq t_r < 1$ . Consider all schedules  $p \rightarrow q$ ,  $p \neq q$ , at time  $t_q$ , and also  $q \rightarrow r$  at time  $t_r$ . Their omitted masses are below  $E$ , and their last terms in (15) equal  $E$ . If all these schedules had error strictly larger than  $E$ , the middle terms would force

$$A_p(t_q) < h_q \quad (p \neq q), \quad A_q(t_q) \leq A_q(t_r) < h_r.$$

Summing gives  $t_q < (n-1)h_q + h_r$ , or

$$(2n-1)E < n-1 - (n-2)m_q - m_r. \quad (17)$$

But  $m_q \geq 1/n$  and  $m_r \geq (1 - m_q)/(n-1)$ . For  $n \geq 3$  the coefficient  $n-2-1/(n-1)$  is positive, so

$$(n-2)m_q + m_r \geq \left(n-2 - \frac{1}{n-1}\right)m_q + \frac{1}{n-1} \geq \frac{n-1}{n}.$$

Equation (17) would imply  $(2n - 1)E < (n - 1)^2/n$ , contrary to the definition of  $E$ . At least one candidate therefore has error at most  $E$ .

For a matching lower bound of  $1/3$ , use three consecutive distinct pure modes of duration  $1/3$ . Every schedule with at most two blocks omits at least one of these modes and incurs its terminal mass. The uniform-input value in (12) supplies the other lower bound. Finally,

$$\frac{(n - 1)^2}{n(2n - 1)} - \frac{1}{3} = \frac{n^2 - 5n + 3}{3n(2n - 1)}$$

is negative at  $n = 3, 4$  and positive for every integer  $n \geq 5$ , giving the claimed transition and extremizers.  $\square$

For an arbitrary horizon  $T > 0$ , set  $E = H_n(T)$  and  $t_i = T - m_i - E$ . The construction requires no search over switching times. Let  $q$  and  $r$  index a largest and second-largest terminal mass. If  $m_q \geq E$ , select  $r \rightarrow q$  at  $\max\{0, T - m_q - E\}$ . Otherwise evaluate  $A(t_q)$  and choose  $p \neq q$  maximizing  $A_p(t_q)$ . If its candidate fails, the  $q \rightarrow r$  candidate at  $t_r$  succeeds. Apart from cumulative integration, this requires  $O(n)$  arithmetic operations and comparisons. The proof covers ties, zero terminal masses, and switching at an endpoint.

**Corollary 3.5** (One-switch grid upper bound). *For every grid  $\mathcal{T}$  and every measurable relaxed input,*

$$\text{OPT}_1^{\mathcal{T}}(A) \leq H_n(T) + \frac{\bar{\Delta}}{2} \quad (n \geq 3).$$

*The coefficient  $H_n(T)/T$  cannot be decreased in a bound whose grid term vanishes under refinement, uniformly over grids and inputs.*

*Proof.* Choose the schedule from Theorem 3.4 and move its switch to a closest grid point. The displacement is at most  $\bar{\Delta}/2$ . The cumulative occupation of either affected mode changes by at most that displacement, so the triangle inequality gives the bound. No new switch is created. For sharpness of the leading coefficient, use the uniform extremizer when  $n \geq 5$ . When  $n = 3, 4$ , use the three pure blocks on successively refined grids containing  $T/3$  and  $2T/3$ . Each input belongs to its grid class, and its grid optimum is at least its continuous optimum  $H_n(T)$ .  $\square$

### 3.4 Exact three-cell minimax and a published conjecture

**Proposition 3.6** (Three unit cells). *For  $n \geq 3$ ,  $F_{n,1}^{\text{unit}}(3) = 2 - 3/n$ .*

*Proof.* For arbitrary relaxed input choose  $q$  with a largest terminal mass and  $p$  with a second largest, and use the three-cell word  $(p, q, q)$ . Every omitted mass is at most 1, because the masses sum to 3. The first-cell error is at most 1. At time 2, the selected modes have occupation 1 and cumulative allocations in  $[0, 2]$ , hence errors at most 1; omitted modes still have errors at most 1. At time 3,  $m_p \leq 3/2$  and its occupation is 1, so its error is at most 1. The final mode satisfies  $3/n \leq m_q \leq 3$  and has occupation 2, giving error at most  $\max\{1, 2 - 3/n\} = 2 - 3/n$ . Endpoint monotonicity proves the upper bound. For uniform input, every schedule with at most one switch gives some mode at least two cells. Its negative terminal discrepancy is at least  $2 - 3/n$ . Thus the upper bound is attained.  $\square$

Sager and Zeile [9, Conjecture 1, equation (7.6)] propose, for  $n > 2$  and  $1 \leq s \leq N - 2$ , the worst-case grid value

$$\begin{cases} T/(s + 2) + \bar{\Delta}/2, & s \leq n - 2, \\ T/(2s + 4 - n) + \bar{\Delta}/2, & s > n - 2. \end{cases} \quad (18)$$

Their initial activation is free, as in our formulation [9, Definition 10]. The following examples satisfy all of the stated restrictions and disprove the first branch even as an upper bound. No conclusion about every case of its second branch is needed here.

**Example 3.7** (Five modes and one switch). Take  $n = 5$ ,  $T = N = 45$  unit cells, and  $\alpha_i = 1/5$  throughout. For any distinct pair of modes and an integer switch time  $\tau$ , the exact error from (15) is

$$D(\tau) = \max\{9, 4\tau/5, 36 - \tau\}.$$

The two nonconstant terms intersect at  $\tau = 20$ , with value 16. For  $\tau \leq 20$  the last term is at least 16; for  $\tau \geq 20$  the middle term is at least 16. Constant schedules are included by the endpoint choices. Thus the grid optimum is 16, whereas (18) gives  $45/3 + 1/2 = 31/2$ . On  $45m$  unit cells the same construction has optimum  $16m$ , exceeding the conjectured  $15m + 1/2$  for every integer  $m \geq 1$ . Equivalently, on a fixed horizon  $T$  with  $45m$  equal cells its optimum is  $16T/45$ , while the conjectured bound tends to  $T/3$ . The failure persists under refinement.

**Example 3.8** (Smallest admissible number of cells). On three unit cells and with seven uniform modes, the exact grid error is  $11/7$  by Proposition 3.6. It exceeds the conjectured  $3/3 + 1/2 = 3/2$ . The restriction  $1 \leq s \leq N - 2$  excludes grids with fewer than three cells, so this witness uses the smallest admissible number of cells.

More generally, for every fixed  $s \geq 1$ , Theorem 3.1 approaches  $T/(s+1) > T/(s+2)$  as the mode count grows. For sufficiently many modes, its continuous uniform value already exceeds the first branch's proposed leading term. A sufficiently fine uniform grid then makes  $\bar{\Delta}/2$  smaller than this strict excess; restricting the uniform input to grid schedules cannot reduce its optimum. Thus this obstruction occurs for every fixed positive switch budget, not only for one switch. In the one-switch case, Theorem 3.4 and Corollary 3.5 give an exact continuous replacement and a valid grid upper bound. The latter is not asserted to be an exact finite-grid equality.

## 4 Heavy modes and the reduction to one-sided error

The two discrepancy signs have different obstructions. An omitted mode contributes its entire terminal mass to  $A - W$ , whereas a long activation block can make  $W - A$  large. The following theorem separates these effects for every block count. A mode is called *heavy at threshold*  $E$  when  $m_i > E$ ; the inequality is strict.

**Theorem 4.1** (Universal heavy-mode rounding). *Let  $s \geq 0$ ,  $E > 0$ , and  $T \leq (s+2)E$ . If some mode has  $m_i > E$ , then there is a schedule with at most  $s$  switches and full error at most  $E$ . There is no restriction relating  $n$  to  $s$ .*

We prove the theorem by rounding prefixes on equal intervals and then reordering a carefully chosen prefix. This reordering is needed because a repeated mode alone does not save a switch: its occurrences must be adjacent.

**Lemma 4.2** (Integral prefix rounding with a repetition). *Let  $A \in \mathcal{A}_n(M)$ , where  $M \geq 2$  is an integer. If  $A_h(M) > 1$ , there is a word  $w_1, \dots, w_M$  on unit intervals whose counts  $N_i(j)$  obey*

$$\lfloor A_i(j) \rfloor \leq N_i(j) \leq \lceil A_i(j) \rceil \quad (i \in [n], 1 \leq j \leq M), \quad (19)$$

*and in which mode  $h$  occurs at least twice. This word has full error at most one for the original measurable input.*

*Proof.* Build a finite network with one supply-one node for each slot  $j$  and, for each mode  $i$ , a chain of nodes  $(i, 1), \dots, (i, M)$ . Slot  $j$  has an arc to each  $(i, j)$ , with capacity one. The chain edge from  $(i, j)$  to  $(i, j + 1)$  has lower and upper capacities  $\lfloor A_i(j) \rfloor$  and  $\lceil A_i(j) \rceil$ ; the last chain edge goes to a common sink with the corresponding capacities at  $M$ . The sink has demand  $M$ . Sending  $A_i(j) - A_i(j - 1)$  from slot  $j$  to its mode chain gives a feasible fractional flow. All bounds and supplies are integers. The integral-flow theorem (equivalently, total unimodularity of a directed incidence matrix) gives an integral optimum for a linear objective on this bounded feasible flow polytope.

Maximize the final flow in chain  $h$ . Its optimum is at least the feasible fractional value  $A_h(M) > 1$ , so an integral optimum assigns at least two slots to  $h$ . Each slot sends its one unit to exactly one mode, and its chain counts satisfy (19). The endpoint error is at most one. Lemma 2.1 extends this bound throughout each unit interval without any assumption that the input is constant there.  $\square$

**Lemma 4.3** (First-repeat reordering). *Any word on  $M$  unit intervals with full error at most one and a repeated mode can be replaced by a word with full error at most one and an equal adjacent pair.*

*Proof.* If there is already an adjacent pair, nothing is required. Otherwise let  $r$  be the first position repeating an earlier mode  $q$ . The prefix of length  $r$  contains  $q$  twice and  $r - 2$  other modes once each. Its error at  $r$  gives

$$1 \leq A_q(r) \leq 3, \quad A_i(r) \leq 2 \text{ for other selected modes}, \quad A_i(r) \leq 1 \text{ for omitted modes.} \quad (20)$$

Only this prefix will be reordered, preserving its occupation multiset.

Let  $d_q = \max\{t \in [0, r] : A_q(t) \leq 1\}$ . Continuity and (20) give  $A_q(d_q) = 1$ , including the case  $A_q(r) = 1$  where  $d_q = r$ . Moreover  $1 \leq d_q \leq r$ , since  $A_q(t) \leq t$ . Choose an integer  $j \in \{0, \dots, r - 2\}$  with  $j \leq d_q \leq j + 2$ ; one choice is  $j = \min\{r - 2, \max\{0, \lceil d_q \rceil - 2\}\}$ . Place  $q$  on the double block  $[j, j + 2]$ .

For another selected mode  $i$  with  $A_i(r) > 1$ , define its deadline  $d_i = \max\{t \leq r : A_i(t) \leq 1\}$ . Give every remaining selected mode the artificial deadline  $r$ . Sort these  $r - 2$  modes by nondecreasing deadline, resolving ties arbitrarily, and assign them to the unit slots with start times

$$0, 1, \dots, j - 1, \quad j + 2, j + 3, \dots, r - 1.$$

We verify that each mode starts by its deadline. If the  $\ell$ th sorted mode precedes  $q$ , it starts at  $\ell - 1$ . A deadline smaller than this would imply that all first  $\ell$  modes have allocation strictly greater than one at  $\ell - 1$ , contradicting total allocation  $\ell - 1$ . If it follows  $q$ , it starts at  $\ell + 1$ , with  $\ell \geq j + 1$ . A deadline smaller than  $\ell + 1$  would give allocation greater than one to the first  $\ell$  modes there. Also  $d_q \leq j + 2 \leq \ell + 1$ , so  $q$  has allocation at least one there. Their combined allocation would exceed  $\ell + 1$ . An artificial deadline  $r$  cannot be violated by a slot start.

For a singly used mode,  $W_i - A_i \leq 1$  because its total service is one. Before activation,  $A_i - W_i \leq 1$  by the deadline property; after activation, its largest positive discrepancy is at  $r$  and is at most  $A_i(r) - 1 \leq 1$ . For  $q$ , the allocation at its block start is at most one and at its block end is at least one. Thus its positive discrepancy at the start and its negative discrepancy at the end are at most one. Its final positive discrepancy is  $A_q(r) - 2 \leq 1$ . Endpoint monotonicity controls the intervening times. Every omitted mode has allocation at most one throughout the prefix. The new prefix therefore has full error at most one. Since its counts at  $r$  are unchanged, attaching the original suffix preserves every later discrepancy. The two  $q$  slots are adjacent.  $\square$

*Proof of Theorem 4.1.* Extend the input from  $T$  to  $(s+2)E$ , when necessary, by assigning all added activity to any mode. An originally heavy mode remains heavy. Scale time and all allocations by  $E$ , giving the integer horizon  $M = s+2$ . Lemmas 4.2 and 4.3 produce an error-one word with an adjacent pair. It has at most  $M-2 = s$  switches. Scale back and restrict to  $[0, T]$ ; neither operation increases the switch count, and the error is at most  $E$ .  $\square$

The reordered word need not preserve the stronger floor/ceiling bounds in (19); the conclusion needed is error at most one. Nor does the proof prescribe the repeated mode: the first repeated mode can differ from the heavy mode used in the flow objective.

**Theorem 4.4** (Exact one-sided reduction). *For every  $1 \leq k < n$ ,*

$$F_{n,k-1}(T) = \max \left\{ \frac{T}{k+1}, G_{n,k}^-(T) \right\}. \quad (21)$$

*Repeated modes are allowed in both optimizations.*

*Proof.* Set  $E = \max\{T/(k+1), G_{n,k}^-(T)\} > 0$ . For an input with a mass greater than  $E$ , Theorem 4.1 applies with  $s = k-1$ . Otherwise every mass is at most  $E$ . Choose a one-sided minimizing schedule, which exists by Proposition 2.2; its negative discrepancy is at most  $G_{n,k}^-(T) \leq E$ . Its positive discrepancy is bounded by  $A_i(t) \leq m_i \leq E$ , regardless of the schedule. This proves the upper bound. Full error dominates one-sided error for every input and schedule, so  $F_{n,k-1} \geq G_{n,k}^-$ . For the remaining lower bound, take  $k+1$  successive distinct pure modes, each for time  $T/(k+1)$ . Every schedule with at most  $k$  blocks omits one of these modes and incurs its full terminal mass as error. The assumption  $k < n$  permits this input.  $\square$

## 5 Analytic reach bounds with two and three distinct modes

Distinct-mode schedules are a useful construction for the one-sided problem, although distinctness is never imposed on the competing optimum. For a threshold  $E > 0$ , a horizon  $L > 0$ , and an unused mode  $i$ , define

$$H_i(t) = t - A_i(t), \quad \Phi_i(b) = \max\{t \in [0, L] : H_i(t) \leq b + E\}, \quad 0 \leq b \leq L. \quad (22)$$

The functions  $H_i$  are continuous and nondecreasing, with  $\sum_i H_i(t) = (n-1)t$ . The feasible set is closed, nonempty, and contains  $b$ ; hence  $\Phi_i(b) \geq b$ . Activating unused mode  $i$  from  $b$  to  $t$  gives endpoint negative discrepancy  $t - b - A_i(t) = H_i(t) - b$ . Its maximum negative discrepancy occurs at the end of that block. Consequently a composition of reach maps along distinct modes constructs a schedule with one-sided error at most  $E$ . Each map is nondecreasing in  $b$ , so using the latest feasible endpoint at each step attains the greatest reach for that word.

We will repeatedly use the exact boundary facts

$$\Phi_i(b) < L \implies H_i(\Phi_i(b)) = b + E, \quad \Phi_i(b) < L \implies H_i(L) > b + E. \quad (23)$$

The first follows by continuity and maximality, and the second because  $L$  is then infeasible. Both remain valid with flat portions of  $H_i$ ; choosing an arbitrary threshold root instead of the latest endpoint would not suffice for all comparisons below. Put

$$r_n = \frac{n}{n-1}, \quad B_j(n, E) = nE(r_n^j - 1), \quad B_0(n, E) = 0. \quad (24)$$

When  $n, E$  are fixed, write  $B_j = B_j(n, E)$ . These values satisfy  $B_j = r_n(B_{j-1} + E)$ .

**Lemma 5.1** (Two distinct blocks). *For  $n \geq 3$ , two distinct modes suffice to reach  $\min\{T, B_2\}$  with one-sided error at most  $E$ . A single mode suffices to reach  $\min\{T, B_1\}$  for every  $n \geq 2$ .*

*Proof.* Define reaches on the target horizon  $L$ . If a candidate already reaches  $L$ , there is nothing to prove. Otherwise the first reaches  $R_i = \Phi_i(0)$  are uncapped, so  $A_i(R_i) = R_i - E$ . For the single-block assertion,  $x = \max_i R_i < L$  implies  $(n-1)x = \sum_i H_i(x) \geq nE$ , so  $x \geq B_1$ , a contradiction when  $L \leq B_1$ .

For two blocks, assume every ordered distinct pair has reach below  $L$ , and let  $M = \max_{p \neq q} \Phi_q(R_p) < L$ . Let  $x \geq y$  be the two largest first reaches and choose  $p$  attaining  $x$ . At time  $y$ ,  $A_p(y) \leq x - E$  and  $A_i(y) \leq y - E$  for every  $i \neq p$ . Summing gives  $x + (n-2)y \geq nE$ . The identity

$$(n-1)x + y = \frac{n}{n-1}[x + (n-2)y] + \frac{n^2 - 3n + 1}{n-1}(x - y) \geq \frac{n^2 E}{n-1} \quad (25)$$

uses  $n \geq 3$ . For each  $i$ , let  $x_i = \max_{j \neq i} R_j$ . Every pair ending in  $i$  reaches no further than  $M$ , so (23) and monotonicity give  $A_i(M) \leq M - E - x_i$ . Sum over  $i$  and note  $\sum_i x_i = (n-1)x + y$ . It follows that

$$(n-1)M \geq nE + (n-1)x + y, \quad M \geq \frac{nE(2n-1)}{(n-1)^2} = B_2. \quad (26)$$

This contradicts  $M < L \leq B_2$  and proves the claim.  $\square$

**Theorem 5.2** (Three distinct blocks). *For  $n \geq 4$ , three distinct modes suffice to reach  $\min\{T, B_3\}$  with one-sided error at most  $E$ .*

*Proof.* Take  $L = \min\{T, B_3\}$  and suppose no sequence of at most three distinct modes reaches  $L$ . In particular every first and pair reach is uncapped. Retain  $R_i, M, x, y$  from Lemma 5.1, and let  $z$  be the third largest first reach. At time  $z$ , the top two modes have allocation at most  $x - E$  and  $y - E$ , and the other modes have allocation at most  $z - E$ . Thus  $x + y + (n-3)z \geq nE$ . Writing  $u = x + y \geq 2z$  gives

$$\begin{aligned} (n-1)u + 2z &= \frac{2n}{n-1}[u + (n-3)z] + \frac{n^2 - 4n + 1}{n-1}(u - 2z) \\ &\geq \frac{2n^2 E}{n-1}, \end{aligned} \quad (27)$$

where the last coefficient is positive for  $n \geq 4$ .

Let  $M_i$  be the largest pair reach excluding mode  $i$ , and let  $x_i \geq y_i$  be the largest two first reaches after that exclusion. For each  $a \neq i$ , all pairs available without  $i$  and ending in  $a$  give

$$A_a(M_i) \leq M_i - E - \max_{b \notin \{i, a\}} R_b.$$

The maxima on the right sum over  $a \neq i$  to  $(n-2)x_i + y_i$ . Summing and using  $\sum_a A_a(M_i) = M_i$  yields

$$A_i(M_i) \geq (n-1)E + (n-2)x_i + y_i - (n-2)M_i.$$

On the other hand,  $M_i \leq M$  and  $A_i(M) \leq M - E - x_i$ . Monotonicity therefore implies

$$(n-2)M_i \geq nE + (n-1)x_i + y_i - M. \quad (28)$$

Choose a pair  $(p, q)$  attaining  $M$ . The same pair is available in every exclusion except possibly  $p, q$ , so  $M_i = M$  outside this pair. Also  $x_p + x_q \geq x + y$  and  $y_p, y_q \geq z$ . Summing (28) for  $p, q$  gives

$$\sum_i M_i \geq \left(n - 2 - \frac{2}{n-2}\right) M + \frac{2nE + (n-1)(x+y) + 2z}{n-2}.$$

The coefficient of  $M$  is positive for  $n \geq 4$ . Substituting  $M \geq B_2$  from (26) and (27) proves

$$\sum_i M_i \geq nB_2. \quad (29)$$

For completeness, the substituted expression equals  $nB_2$  because  $(n-1)B_2 = nE + n^2E/(n-1)$ .

Appending mode  $i$  to a maximizing pair excluding it fails to reach  $L$  by assumption. The second part of (23) gives  $H_i(L) > M_i + E$ . Sum these strict inequalities and use (29):

$$(n-1)L > \sum_i M_i + nE \geq nB_2 + nE = (n-1)B_3.$$

This contradicts  $L \leq B_3$ . □

This proof is constructive: compute the first reaches and the ordered-pair reach matrix, then append each possible final mode to a maximizing pair excluding it. A global maximizing pair supplies all but at most two of these excluded maxima; the remaining two require additional scans of the matrix. The construction therefore uses  $n(n-1)$  pair reach evaluations, followed by  $n$  final reach evaluations and  $O(n^2)$  comparisons. It replaces an earlier proof for  $n = 4$  based on thirty event-order linear programs. Those exact certificates are included as an independent special-case check in the supplement, but are not premises of Theorem 5.2.

## 6 Four distinct blocks: an exact certificate proof

The next reach bound has the same uniform-input constant, but its proof uses a finite family of exact linear identities. We give the complete necessary-condition relaxation and its symmetry reduction before applying the certificates. The computer-assisted part establishes only the weighted pair inequality below; the passage from that inequality to a four-block schedule is analytic. No time discretization of the input is involved.

**Theorem 6.1** (Four distinct blocks, computer-assisted). *For  $n \geq 5$ , four distinct modes suffice to reach  $\min\{T, B_4(n, E)\}$  with one-sided error at most  $E$ .*

### 6.1 The weighted pair inequality

Use the reach maps (22) on  $[0, L]$ . Let  $R_i = \Phi_i(0)$  and define the pair maxima

$$\begin{aligned} M &= \max_{a \neq b} \Phi_b(R_a), \\ P_i &= \max_{a, b \notin \{i\}, a \neq b} \Phi_b(R_a), \\ Q_{ij} = Q_{ji} &= \max_{a, b \notin \{i, j\}, a \neq b} \Phi_b(R_a), \quad i \neq j. \end{aligned} \quad (30)$$

These sets of pairs are nonempty. Assume  $M < L$ , so all pair and first reaches are uncapped. For a three-element subset  $S \subset [n]$ , write

$$\mathcal{H}_S = \sum_{i \in S} \left( P_i + \sum_{j \neq i} Q_{ij} \right). \quad (31)$$

An excluded pair wholly in  $S$  is counted twice in this sum.

**Lemma 6.2** (Weighted pair inequality, computer-assisted). *For  $n \geq 5$ ,  $M < L$ , any three-element set  $S$ , and any distinguished mode  $z$ ,*

$$\mathcal{H}_S \geq \frac{3n^2}{n-1}E + \frac{3n^2}{(n-1)^2} \sum_{i \neq z} R_i. \quad (32)$$

In particular,

$$\mathcal{H}_S \geq 3nB_2(n, E). \quad (33)$$

To see the implication between the two displayed inequalities, choose  $z$  with smallest first reach. Then

$$R_z = \sum_i A_i(R_z) \leq \sum_i A_i(R_i) = \sum_i (R_i - E), \quad \sum_{i \neq z} R_i \geq nE.$$

Substitution in (32) gives (33), since  $B_2 = nE(2n-1)/(n-1)^2$ . The stronger statement is what the certificates prove, and does not assume that  $z$  is smallest.

## 6.2 The complete necessary-condition linear program

Scale time by  $E$ , so that  $E = 1$ . For this subsection and its implementation only, label modes  $0, \dots, n-1$  and set  $S = \{0, 1, 2\}$ . The variables are nonnegative  $R_i$ , and event times  $t_v$  and allocations  $a_{v,i}$  for

$$v \in \mathcal{V} = \{M\} \cup \{P_i : i \in S\} \cup \{Q_{ij} : i < j, \{i, j\} \cap S \neq \emptyset\}.$$

There are  $1 + 3 + [3(n-3) + 3] = 3n - 2$  events. For event  $v$ , let  $D_v$  be its excluded set:  $\emptyset$ ,  $\{i\}$ , or  $\{i, j\}$  as appropriate. Choose an unordered pair  $P = \{p, q\}$  attaining the global pair maximum  $M$ . The relaxation consists of exactly the following constraints:

$$\sum_{i=0}^{n-1} a_{v,i} = t_v \quad (v \in \mathcal{V}), \quad (34)$$

$$R_j - t_v + a_{v,k} \leq -1 \quad (v \in \mathcal{V}; j, k \notin D_v, j \neq k), \quad (35)$$

$$a_{Q_{ij},k} \leq a_{P_i,k} \quad (i \in S, j \neq i; 0 \leq k < n), \quad (36)$$

$$a_{v,k} \leq a_{M,k} \quad (v \in \mathcal{V} \setminus \{M\}; 0 \leq k < n), \quad (37)$$

$$a_{M,k} \leq a_{Q_{ij},k} \quad (\{i, j\} \cap P = \emptyset, Q_{ij} \in \mathcal{V}; 0 \leq k < n), \quad (38)$$

$$a_{M,k} \leq a_{P_i,k} \quad (i \in S \setminus P; 0 \leq k < n). \quad (39)$$

All variable nonnegativity constraints are imposed as well. The objective is

$$C = (n-1)^2 \left( \sum_{i \in S} t_{P_i} + \sum_{\substack{i < j \\ \{i, j\} \cap S \neq \emptyset}} |\{i, j\} \cap S| t_{Q_{ij}} \right) - 3n^2 \sum_{i \neq z} R_i. \quad (40)$$

We will certify  $C \geq 3n^2(n-1)$ .

Every input in the uncapped case gives a feasible point: set  $t_v$  to its pair reach and  $a_{v,i} = A_i(t_v)$ . Equation (34) expresses total allocation. For (35), the event reach is at least  $\Phi_k(R_j)$  and this endpoint is uncapped. Hence  $H_k(t_v) \geq R_j + 1$ . Excluding more modes cannot increase a pair maximum, which gives (36) and (37) by allocation monotonicity. If an exclusion avoids the maximizing pair  $P$ , that pair remains available and its event time

| $P$        | Representatives for $z$ | Range      |
|------------|-------------------------|------------|
| $\{0, 1\}$ | $0, 2, 3$               | $n \geq 5$ |
| $\{0, 3\}$ | $0, 1, 3, 4$            | $n \geq 5$ |
| $\{3, 4\}$ | $0, 3$                  | $n \geq 5$ |
| $\{3, 4\}$ | $5$                     | $n \geq 6$ |

Table 2: All ten types of  $(S, P, z)$  after relabeling  $S = \{0, 1, 2\}$ .

equals  $M$ . The reverse allocation inequalities (38)–(39) express these equalities. Together with (34), the two allocation orders also force the event times to be equal.

There are no additional constraints: in particular we impose neither first-root equations nor an ordering of the  $R_i$ , and no chronological ordering between unrelated events. The relaxation can therefore include nonphysical points. A lower bound valid on this larger feasible set is nevertheless valid for every control. No converse representation claim is used.

### 6.3 Exhaustive symmetry reduction

Permutations preserving  $S$  carry a maximizing pair to one of three representatives, according to its intersection size with  $S$ . Permutations preserving both  $S$  and this pair then place  $z$  in the representatives shown in Table 2. For example, when  $P = \{0, 3\}$ , the four possible positions of  $z$  are the singleton  $S \cap P$ , one of the two elements of  $S \setminus P$ , the singleton  $P \setminus S$ , and an element outside  $S \cup P$ . These are exactly the four displayed cases. The same partition argument proves the other two rows. Ties between maximizing pairs do not matter: any maximizing pair may be chosen.

For one type, keep the indices in  $K = S \cup P \cup \{z\}$  individually labeled and let the remaining indices be interchangeable. Averaging a feasible point over all permutations of those remaining indices preserves every constraint and the objective. Thus restricting variables to be constant on permutation orbits preserves the infimum of the relaxation. This assertion needs no optimizer or boundedness assumption: every feasible point has an invariant feasible average with the same value.

Here is an explicit orbit rule sufficient to reconstruct the quotient. A variable is identified by  $(R, i)$ ,  $(t, v)$ , or  $(a, v, i)$ . Retain each index in  $K$  verbatim. Replace each other index, in order of first occurrence in this identifier, by  $b_0, b_1, \dots$ , reusing its symbol on subsequent occurrences. A  $Q$  event contains at most one interchangeable index, because its excluded pair meets  $S$ . Thus this rule preserves exactly the equalities and inequalities of the unspecified indices, without an orientation ambiguity for its unordered pair. In particular, allocation to the interchangeable index excluded by a  $Q$  event and allocation to a different interchangeable index remain separate variables.

Replace variables in each row of (34)–(39) by these orbit labels, add coefficients that now refer to the same variable, and delete duplicate rows. In the objective, add all multiplicities of the same variable orbit. This defines the quotient matrices and objective uniquely, independently of any numerical solver. The bundled checker implements precisely this construction. Its ordering enumerates  $R_i$  first, then  $M$ , the  $Q_{ij}$  in lexicographic order, and the three  $P_i$ ; it retains the first occurrence of each variable orbit and each distinct row. The certificate coordinates refer to that ordering.

At most six indices lie in  $K$ . A pair-constraint row involves at most three interchangeable indices: one from its event and its two available modes. An allocation-order row involves at most two. Thus all variable and row patterns are already present once  $n \geq 9$ . Additional interchangeable indices repeat those patterns. They affect only multiplicities in mass

equations and the objective. If  $a = |K|$ , a mass row has  $n - a$  identical allocation terms, or one separately labeled term and  $n - a - 1$  others when an interchangeable index is excluded by the event. Hence its coefficients are affine functions of  $n$ . Objective orbit multiplicities are constant or  $n - a$ , multiplied by  $(n - 1)^2$  for event times or by  $-3n^2$  for first reaches. All objective coefficients are consequently integer polynomials of degree at most three. The inequality matrix and right-hand sides are constant for  $n \geq 9$ .

For  $n \geq 9$ , the quotient has between 57 and 162 variables, 151 and 826 inequality rows, and 10 and 19 equality rows, depending on the type. For the symbolic check, the exact affine mass coefficients and exact affine objective multiplicities are reconstructed from the quotient at  $n = 9, 10$ . This is reconstruction of the established affine counts, not interpolation of an unknown function. All individually labeled indices are at most 5, and three interchangeable indices suffice for every row pattern. First-occurrence ordering of patterns is therefore stable when further indices are added.

#### 6.4 Exact dual identities and the verification boundary

Write the quotient as  $Ax \leq b$ ,  $B(n)x = d$ ,  $x \geq 0$ , with objective  $c(n)^T x = C$ . In this notation  $b$  has entries zero or minus one and  $d = 0$ . For a finite dimension, a certificate supplies a positive integer denominator  $D$  and integer vectors  $Y \leq 0$ ,  $Z$  satisfying

$$Dc = A^T Y + B^T Z, \quad b^T Y + d^T Z = 3n^2(n - 1)D. \quad (41)$$

For every feasible  $x$ , multiplying its inequalities by the nonpositive entries of  $Y$  reverses their direction. Therefore

$$DC = Y^T Ax + Z^T Bx \geq Y^T b + Z^T d = 3n^2(n - 1)D.$$

No dual optimality assertion is required. The certificates have zero coefficient residual, so nonnegative-variable residual multipliers are unnecessary.

For  $5 \leq n \leq 22$ , the data contain all nine types at  $n = 5$  and all ten at each  $n = 6, \dots, 22$ : a total of  $9 + 17 \cdot 10 = 179$  finite certificates. The checker verifies the case identifiers, signs, integrality, every coordinate of (41), and the stated lower bound.

For each type, a further certificate gives integer polynomials  $D(n)$ ,  $D_0(n)$ ,  $Y(n)$ , and  $Z(n)$  satisfying

$$D(n) = (n - 1)^2 D_0(n), \quad (42)$$

$$D_0(n)c(n) = A^T Y(n) + B(n)^T Z(n), \quad (43)$$

$$b^T Y(n) + d^T Z(n) = 3n^2(n - 1)D_0(n). \quad (44)$$

The checker verifies these coefficient by coefficient, using integer polynomial addition and multiplication. It expands  $D(23 + t)$  and each entry of  $-Y(23 + t)$ . The former has positive constant coefficient and nonnegative remaining coefficients; the latter have only nonnegative coefficients. Consequently  $D(n) > 0$  and  $Y(n) \leq 0$  for every real  $n \geq 23$ , and (42) gives  $D_0(n) > 0$  there. Dividing (43)–(44) by  $D_0(n)$  gives the same dual bound for every integer  $n \geq 23$ . These checks cover all dimensions  $n \geq 5$  and prove (32).

The supplement's `verification/reference/` directory contains the complete data in `certificates_general_four_block.json` and the checker `verify_general_four_block.py`. Run, from the paper directory,

```
python verification/reference/verify_general_four_block.py
```

The checker requires only the Python standard library and uses integer arithmetic throughout. It reads certificate arrays as data, not executable expressions. Numerical optimization and symbolic algebra were used to discover the certificates; neither a numerical solution, a floating-point tolerance, nor an optimization status is trusted by the checker. Its successful output verifies all 179 finite cases and all ten polynomial cases. A provenance manifest records SHA-256 hashes of the bundled original artifacts. The proof's computational dependency is this finite data and its exact verification, together with the mathematical reduction and orbit coverage above. It is explicitly a computer-assisted proof.

## 6.5 From weighted pairs to four distinct blocks

*Proof of Theorem 6.1.* Put  $L = \min\{T, B_4\}$  and assume no sequence of at most four distinct modes reaches  $L$ . All first, pair, and triple reaches are then uncapped. Let  $G$  be the maximum three-distinct-mode reach and  $N_i$  the corresponding maximum excluding  $i$ . Theorem 5.2 implies  $G \geq B_3$ : if  $L \leq B_3$ , that theorem already contradicts the assumed failure.

For each  $k \neq i$ , appending  $k$  to a pair attaining  $Q_{ik}$  gives a triple excluding  $i$  and ending no later than  $N_i$ . Hence  $A_k(N_i) \leq N_i - E - Q_{ik}$ . Summing over  $k \neq i$  gives

$$A_i(N_i) \geq (n-1)E + \sum_{k \neq i} Q_{ik} - (n-2)N_i.$$

Appending  $i$  to a pair attaining  $P_i$  similarly gives  $A_i(G) \leq G - E - P_i$ . As  $N_i \leq G$ , allocation monotonicity implies

$$(n-2)N_i \geq nE + P_i + \sum_{k \neq i} Q_{ik} - G. \quad (45)$$

Choose a triple attaining  $G$  and let  $S$  be its mode set. Then  $N_i = G$  outside  $S$ . Sum (45) over its three elements:

$$\sum_i N_i \geq \left(n-3 - \frac{3}{n-2}\right)G + \frac{3nE + \mathcal{H}_S}{n-2}.$$

The coefficient of  $G$  is positive for  $n \geq 5$ . Applying  $G \geq B_3$  and Lemma 6.2 yields

$$\sum_i N_i \geq nB_3, \quad (46)$$

because  $n(B_2 + E) = (n-1)B_3$ .

Finally, appending  $i$  to a maximizing triple excluding it uses four distinct modes. Its failure implies  $H_i(L) > N_i + E$ . Summing gives

$$(n-1)L > \sum_i N_i + nE \geq nB_3 + nE = (n-1)B_4,$$

contradicting  $L \leq B_4$ . The strict inequality follows from latest endpoints even when  $H_i$  has flat portions. Endpoint monotonicity makes the endpoint constraints control the entire one-sided error trajectory; unselected modes have  $W_i - A_i = -A_i \leq 0$ .  $\square$

An earlier five-mode proof used a separate weighted-pair event relaxation. Its verifier and certificate data are retained in the supplement as an independent check of the smallest case. The all-dimension certificates above subsume that positive result. Neither the five-mode checker nor the thirty-case three-block checker is needed to establish the theorem as presented here.

## 7 Exact minimax values through three switches

The analytic reach bounds and the four-block certificate theorem determine both the one-sided minimax and the full minimax. A common formulation also preserves the stronger conclusion available under equal terminal masses.

**Theorem 7.1** (One-sided minimax through four blocks). *For  $1 \leq k \leq 4$  and  $n \geq k + 1$ ,*

$$G_{n,k}^-(T) = \frac{T}{n(r_n^k - 1)}. \quad (47)$$

*The constant uniform input attains this value. For  $k = 4$  the upper bound uses the computer-assisted Theorem 6.1; for  $k \leq 3$  its proof is analytic.*

*Proof.* Use the single-, two-, three-, or four-block reach result, respectively, at  $E = T/[n(r_n^k - 1)]$ . It reaches  $T$  and proves the upper bound with distinct modes. For the uniform input  $A_i(t) = t/n$ , let an arbitrary schedule, allowing repetitions, have one-sided error  $E$  and block endpoints  $t_j$ . At the end of block  $j$ , the active mode's occupation is at least  $t_j - t_{j-1}$ , so

$$t_j - t_{j-1} - t_j/n \leq E, \quad t_j \leq r_n(t_{j-1} + E).$$

Iteration yields  $T \leq nE(r_n^k - 1)$ . Shorter schedules can be padded with zero-length blocks, or the recurrence can be iterated further using its monotonicity. This proves the matching lower bound even when modes repeat.  $\square$

**Theorem 7.2** (Full minimax through three switches). *For  $0 \leq s \leq 3$  and  $n \geq s + 2$ ,*

$$F_{n,s}(T) = T \max \left\{ \frac{1}{s+2}, \frac{1}{n(r_n^{s+1} - 1)} \right\}. \quad (48)$$

*In particular,*

$$F_{n,2}(T) = T \max \left\{ \frac{1}{4}, \frac{(n-1)^3}{n(3n^2 - 3n + 1)} \right\}, \quad n \geq 4, \quad (49)$$

$$F_{n,3}(T) = T \max \left\{ \frac{1}{5}, \frac{(n-1)^4}{n(4n^3 - 6n^2 + 4n - 1)} \right\}, \quad n \geq 5. \quad (50)$$

*The three-switch upper bound is computer-assisted; the other cases are analytic.*

*Proof.* Combine Theorems 4.4 and 7.1. The two matching input types are the  $s+2$  successive distinct pure modes of equal duration and the constant uniform input. Theorem 3.1 gives the latter's full error, whose additional  $T/n$  term is bounded by  $T/(s+2)$  in this range.  $\square$

**Corollary 7.3** (Equal terminal masses). *Let  $1 \leq k \leq 4$  and  $n \geq k + 1$ . Over inputs with  $m_i = T/n$  for every mode, the exact worst-case full error with at most  $k$  blocks is*

$$\max_{\substack{A \in \mathcal{A}_n(T) \\ A_i(T) = T/n \ (i \in [n])}} \text{OPT}_{k-1}(A) = T \max \left\{ \frac{1}{n}, \frac{1}{n(r_n^k - 1)} \right\}. \quad (51)$$

*The constant uniform input is an extremizer in this restricted class.*

*Proof.* The reach theorem controls  $W - A$  at the second threshold in (51). Independently, every positive discrepancy is bounded by  $A_i(t) \leq m_i = T/n$ . Combining these two bounds proves the upper bound. The uniform input belongs to the class and attains the displayed value by Theorem 3.1.  $\square$

The equal-mass conclusion is stronger than merely applying the unrestricted minimax bound to these inputs. For two switches it gives  $T/4$  at  $n = 4$  and the uniform recurrence value for all  $n \geq 5$ . Thus at  $n = 5, 6, 7$  the restricted worst case is strictly smaller than the unrestricted value  $T/4$ . For three switches it gives  $T/n$  at  $n = 5, 6$  and the uniform recurrence value for  $n \geq 7$ ; it is strictly smaller than  $T/5$  for  $n = 6, \dots, 11$ . The common corollary derives the equal-mass result for four blocks as well as the earlier analytic three-block result without needing a separate construction for each mass-constrained class.

## 7.1 Plateau transitions and mode-count asymptotics

For two switches, the uniform coefficient in (49) exceeds  $1/4$  exactly when

$$p_2(n) = n^3 - 9n^2 + 11n - 4 > 0.$$

Direct substitution gives negative values at  $n = 4, 5, 6, 7$ . At  $n = 8 + h$ ,

$$p_2(8 + h) = h^3 + 15h^2 + 59h + 20 > 0 \quad (h \geq 0).$$

Thus  $F_{n,2}(T) = T/4$  for  $4 \leq n \leq 7$ , and uniform controls are worst-case for all  $n \geq 8$ .

For three switches, the corresponding sign polynomial is

$$p_3(n) = n^4 - 14n^3 + 26n^2 - 19n + 5.$$

It is negative at  $n = 5, \dots, 11$ , whereas

$$p_3(12 + h) = h^4 + 34h^3 + 386h^2 + 1469h + 65 > 0 \quad (h \geq 0).$$

Hence  $F_{n,3}(T) = T/5$  for  $5 \leq n \leq 11$ , and uniform controls are worst-case for  $n \geq 12$ .

Expanding the exact rational expressions gives

$$F_{n,2}(T) = \frac{T}{3} - \frac{2T}{3n} + \frac{2T}{9n^2} + O(Tn^{-3}), \quad (52)$$

$$F_{n,3}(T) = \frac{T}{4} - \frac{5T}{8n} + \frac{5T}{16n^2} + O(Tn^{-3}). \quad (53)$$

These are expansions of the full minimax, since the uniform term dominates for all sufficiently large  $n$ , rather than lower-bound expansions alone.

## 8 Structural limits of the constructions

The hypotheses and choices in the preceding arguments matter. We give exact counterexamples to three tempting extensions. In the first example, the failure extends beyond distinct-mode schedules: allowing a repeated mode still does not restore uniform extremality at the boundary  $k = n$ . This strengthens what a failure of a distinct-mode construction alone would imply.

### 8.1 A three-mode obstruction with an exact instance optimum

**Proposition 8.1** (Failure at  $k = n = 3$ ). *There is an input  $A \in \mathcal{A}_3(57/8)$  with*

$$\text{OPT}_3^-(A) = \frac{18673}{18396} > 1, \quad (54)$$

*although the constant uniform input on that horizon has one-sided optimum exactly one with three blocks. An optimal word is  $(0, 2, 1)$ , with switching times  $8341/4599$  and  $17639/4599$ . Moreover, every three-distinct-mode reach composition at threshold one stops at  $971/146 < 57/8$  for this input. In particular,  $G_{3,3}^-(57/8) \geq 18673/18396 > 1$ .*

| $146t$ | $146A_0(t)$ | $146A_1(t)$ | $146A_2(t)$ |
|--------|-------------|-------------|-------------|
| 0      | 0           | 0           | 0           |
| 146    | 98          | 48          | 0           |
| 194    | 110         | 48          | 36          |
| 256    | 110         | 78          | 68          |
| 408    | 224         | 116         | 68          |
| 516    | 224         | 178         | 114         |
| 580    | 240         | 178         | 162         |
| 971    | 417         | 309         | 245         |

Table 3: Rational cumulative-allocation knots for the three-mode example. Every entry, including time, is divided by 146.

*Proof.* For this example label the modes 0, 1, 2. Define their cumulative allocations by linear interpolation through Table 3, dividing every entry by 146. After its last knot  $t_* = 971/146$ , extend with rates  $(1/3, 1/3, 1/3)$  to  $L = 57/8$ . Every allocation increment is nonnegative and the three increments sum to its time increment. Each individual slope is at most  $3/4$ , so this is an admissible control and every  $H_i(t) = t - A_i(t)$  is strictly increasing.

At threshold  $E = 1$  the first reaches are

$$(R_0, R_1, R_2) = (128/73, 97/73, 1).$$

Both orders of the other two modes after excluding  $i$  have the same pair reach, respectively

$$(M_0, M_1, M_2) = (204/73, 258/73, 290/73).$$

These identities follow by substituting the relevant knots into  $H_i(R_i) = 1$  and  $H_k(M_i) = R_j + 1$  for distinct  $i, j, k$ . At the last knot,  $H_i(t_*) = M_i + 1$  for each  $i$ . Strict increase proves that all six three-distinct-mode compositions end exactly at  $t_*$ .

We first bound distinct-mode schedules at thresholds slightly above one. Write

$$\delta_* = \frac{277}{18396}, \quad E_* = 1 + \delta_* = \frac{18673}{18396}, \quad L - t_* = \frac{63}{2}\delta_*.$$

For  $y \geq 0$ , define the capped inverse  $\psi_i(y) = \max\{t \in [0, L] : H_i(t) \leq y\}$ , so the threshold- $E$  reach map is  $\Phi_i^E(b) = \psi_i(b + E)$ . The slope bound gives  $H_i(t) - H_i(u) \geq (t - u)/4$  for  $u \leq t$ . Consequently  $\psi_i$  is nondecreasing and 4-Lipschitz. Indeed, if  $\psi_i(y) < \psi_i(z)$ , the lower endpoint is uncapped and satisfies  $H_i(\psi_i(y)) = y$ , whereas  $H_i(\psi_i(z)) \leq z$ , even if the upper endpoint is  $L$ . This proves  $\psi_i(z) - \psi_i(y) \leq 4(z - y)$  without excluding caps.

For  $E = 1 + \delta$  with  $0 \leq \delta \leq \delta_*$ , the first reach  $R_p(E) = \Phi_p^E(0)$  is at most  $R_p + 4\delta$ . A pair using the two modes other than  $i$ , in either order, therefore reaches at most  $M_i + 4(4\delta + \delta) = M_i + 20\delta$ . On the uniform suffix,

$$H_i(L) = M_i + 1 + \frac{2}{3}(L - t_*) = M_i + 1 + 21\delta_*. \quad (55)$$

For the last mode  $i$  to reach  $L$ , its previous endpoint  $b$  must satisfy  $H_i(L) \leq b + E \leq M_i + 1 + 21\delta$ . This is impossible when  $\delta < \delta_*$ . Equivalently, every distinct triple reaches at most  $\min\{L, t_* + (63/2)\delta\}$ , since its last inverse starts at  $t_*$  and has slope  $3/2$  on the uniform suffix.

The word  $(0, 2, 1)$  attains all these bounds at  $\delta = \delta_*$ . In fact,  $H_0$  has slope  $1/4$  on  $[256/146, 408/146]$ , starting at  $R_0$ ;  $H_2$  has slope  $1/4$  on  $[516/146, 580/146]$ , starting at  $M_1$ . The times

$$u_* = R_0 + 4\delta_* = \frac{8341}{4599}, \quad v_* = M_1 + 20\delta_* = \frac{17639}{4599} \quad (56)$$

lie in these respective intervals. Thus  $H_0(u_*) = E_*$  and  $H_2(v_*) = u_* + E_*$ . Equation (55) gives  $H_1(L) = v_* + E_*$ . Each active mode has negative discrepancy exactly  $E_*$  at its block end. Endpoint monotonicity proves that this schedule has one-sided error  $E_*$ .

We must still exclude repeated-mode schedules below this value. The terminal masses are

$$(m_0, m_1, m_2) = \frac{1}{1752}(5281, 3985, 3217), \quad m_0 > 2E_*, \quad m_1 > 2E_*. \quad (57)$$

For a schedule with active-mode set  $J$  and one-sided error at most  $E_*$ , its terminal inequalities imply

$$L = \sum_{i \in J} W_i(L) \leq \sum_{i \in J} (m_i + E_*), \quad \sum_{i \notin J} m_i \leq |J|E_*.$$

If at most two modes are used, neither mode 0 nor mode 1 can be omitted. The only possible active set is  $\{0, 1\}$ . Every schedule on this set with at most three blocks has the form  $p, q, p$ , with  $(p, q) = (0, 1)$  or  $(1, 0)$ , allowing zero-length blocks to include shorter schedules.

Let  $u \leq v$  be its first two endpoints. Since  $q$  has not been used before its middle block, their constraints give  $u \leq R_p(E_*)$  and  $v \leq \Phi_q^{E_*}(u) \leq M_2 + 20\delta_*$ . The latter upper bound is below  $L$ . The allocation slope bound  $3/4$  therefore yields

$$v - u \leq A_q(v) + E_* \leq A_q(M_2) + 15\delta_* + 1 + \delta_* = M_2 - R_p + 16\delta_*. \quad (58)$$

Here  $A_q(M_2) + 1 = M_2 - R_p$  follows from the threshold-one pair identity. The final occupation of  $p$  is  $L - (v - u)$ , so its terminal constraint requires  $v - u \geq L - m_p - 1 - \delta_*$ . For  $p = 0$  and  $p = 1$ , respectively, the necessary lower bound minus the upper bound in (58) is

$$\frac{781}{876} - 17\delta_* = \frac{2923}{4599} > 0, \quad \frac{1057}{876} - 17\delta_* = \frac{4372}{4599} > 0.$$

Thus no repeated-mode schedule is feasible even at  $E_*$ . Feasibility is monotone in the error threshold, so none is feasible below  $E_*$ . The distinct-mode argument excludes every threshold in  $[1, E_*)$ ; infeasibility at one also excludes smaller thresholds. Together with the matching schedule, this proves (54).

Finally, the uniform input has the distinct-mode schedule with endpoints  $3/2, 15/4, 57/8$ ; each active mode has negative discrepancy one at the end of its block. Conversely, the block-end recurrence  $t_j \leq (3/2)(t_{j-1} + E)$ , valid even with repeated modes, gives  $57/8 \leq 3E((3/2)^3 - 1) = (57/8)E$ . Thus its one-sided optimum is exactly one, as asserted.  $\square$

By scaling, the example gives

$$G_{3,3}^-(T) \geq \frac{37346}{262143}T > \frac{8}{57}T,$$

where the last coefficient is the uniform-input one-sided optimum per unit horizon. This is a lower bound on the minimax, not a determination of its exact value. The supplement checks the knot slopes, all six threshold-one reach compositions, inverse-sensitivity arithmetic, matching schedule, terminal masses, and both middle-service contradictions using rational arithmetic. It also independently rules out every three-block word at threshold one by exact switching-time polygon feasibility. This counterexample does not concern the spare-mode range  $k < n$  of Theorems 4.4 and 7.1, nor does it have equal terminal masses.

## 8.2 A specified heavy mode need not admit an adjacent pair

Theorem 4.1 chooses a repeat; it does not promise to repeat any prescribed heavy mode. On four unit intervals, consider the interval rates

| mode | 1    | 2   | 3 | 4   | total |
|------|------|-----|---|-----|-------|
| 0    | 7/15 | 0   | 0 | 0   | 7/15  |
| 1    | 2/15 | 1/2 | 0 | 4/5 | 43/30 |
| 2    | 2/5  | 1/2 | 1 | 1/5 | 21/10 |

Mode 1 is heavy at threshold one. If an error-one unit-slot word selected it in slots 1–2 or 2–3, its occupation at the pair’s end would be at least two, but its allocation is only 19/30. This gives negative error at least  $41/30 > 1$ . If its pair occupied slots 3–4, mode 2 would need at least two selections in slots 1–2, because its terminal mass is  $21/10 > 2$ . Yet two such selections give error  $2 - 9/10 = 11/10 > 1$  at prefix 2. No location of the prescribed pair works. The words  $(0, 2, 2, 1)$  and  $(0, 1, 2, 2)$  both have error at most one, consistently with the existential theorem. This statement concerns rounding on the fixed unit slots; it does not impose a restriction on arbitrary continuous block lengths.

## 8.3 The pair position cannot be fixed by the first unit-mass prefix

Even if a largest-mass mode is chosen, the following rule can fail: force its two adjacent slots to end at the first integer prefix where its allocation reaches one. On five unit intervals take

| mode | 1    | 2   | 3     | 4   | 5 | total |
|------|------|-----|-------|-----|---|-------|
| 0    | 3/5  | 0   | 1/10  | 2/5 | 1 | 21/10 |
| 1    | 3/10 | 3/5 | 11/20 | 3/5 | 0 | 41/20 |
| 2    | 1/10 | 2/5 | 7/20  | 0   | 0 | 17/20 |

Mode 0 has strictly largest mass and first reaches allocation at least one at prefix 4. Forcing it in slots 3–4 leaves mode 1 with no service there. At prefix 4, mode 1 already has allocation  $41/20 > 2$ , so an error-one word must have selected it in both slots 1–2. Its allocation at prefix 2 is only 9/10, giving the impossible negative error 11/10. In contrast,  $(0, 1, 1, 0, 0)$  has error at most one and repeats the largest mode in slots 4–5. Thus both the selected mode and the pair position are meaningful choices in the prefix-reordering proof. These examples do not disprove the separate conjecture that some error-one word can always repeat a largest heavy mode adjacently; that strengthening is not used by any theorem here.

## 9 General budgets by removing a mode

The exact reach results above provide useful seeds, but a bound for every block budget need not assume any higher-order reach conjecture. We first prove a transfer principle for arbitrary measurable inputs. It preserves distinctness of the selected modes and works even when the one-sided coefficient is smaller than the reciprocal number of modes.

**Lemma 9.1** (Mode removal). *Let  $n \geq 3$  and  $k \geq 2$ . Suppose that, on every positive horizon, every  $(n - 1)$ -mode input admits at most  $k - 1$  distinct activation blocks with one-sided error at most  $C$  times its horizon, where  $C > 0$ . Then the corresponding  $n$ -mode,  $k$ -block coefficient is*

$$f_n(C) = \frac{(n - 1)^2 C + 1}{n(n - 1)(1 + C)}. \quad (59)$$

One may remove a mode of largest terminal mass if  $C \geq 1/(n-1)$ , or one of smallest terminal mass if  $C \leq 1/(n-1)$ . At equality either choice is valid.

*Proof.* Set  $E = f_n(C)T > 0$ , choose the indicated mode  $q$ , and write  $a = A_q(T)$ . If  $a \geq T - E$ , use  $q$  throughout. Its one-sided discrepancy is  $t - A_q(t)$ , which is nondecreasing and ends at  $T - a \leq E$ ; every other mode has nonpositive discrepancy  $W_i - A_i$ . This branch uses one block. Otherwise put

$$L = T - a - E > 0, \quad E' = E - \frac{a}{n-1}. \quad (60)$$

We verify both the positivity of  $E'$  and the inequality  $CL \leq E'$ . A useful identity is

$$f_n(C) - \frac{1}{n} = \frac{(n-2)((n-1)C-1)}{n(n-1)(1+C)}. \quad (61)$$

If  $C \geq 1/(n-1)$ , then  $E \geq T/n$  and  $a < T - E \leq (n-1)E$ , so  $E' > 0$ . Moreover  $a \geq T/n$  and the function

$$a \mapsto \frac{CT + (1/(n-1) - C)a}{1+C}$$

is nonincreasing. Its value at  $T/n$  is exactly  $E$ , whence

$$E \geq \frac{CT + (1/(n-1) - C)a}{1+C} \iff CL \leq E'. \quad (62)$$

If  $0 < C < 1/(n-1)$ , then  $a \leq T/n$  and

$$\frac{T}{n(n-1)} < E < \frac{T}{n}.$$

The first strict inequality follows by subtracting  $1/[n(n-1)]$  from  $f_n(C)$ ; the numerator of the difference is  $n(n-2)C > 0$  before cancelling the common denominator. Thus  $E' > 0$ , and also  $L > T(1-2/n) > 0$ , so the constant-mode branch cannot occur. The same function of  $a$  is now increasing, and  $a \leq T/n$  again proves (62). At  $C = 1/(n-1)$  the function is constant; for minimum-mass selection  $a \leq T/n < (n-1)E$  also verifies positivity.

For  $i \neq q$  complete the remaining rates on  $[0, L]$  by

$$\hat{\alpha}_i(t) = \alpha_i(t) + \frac{\alpha_q(t)}{n-1}, \quad \hat{A}_i(t) = A_i(t) + \frac{A_q(t)}{n-1}.$$

They are measurable, nonnegative, and sum to one. Apply the assumed  $(k-1)$ -block construction to this input on  $[0, L]$ . Its error against the original cumulative allocations satisfies, for every  $t \leq L$ ,

$$W_i(t) - A_i(t) = W_i(t) - \hat{A}_i(t) + \frac{A_q(t)}{n-1} \leq CL + \frac{a}{n-1} \leq E.$$

Mode  $q$  has no service on this prefix and therefore has nonpositive one-sided discrepancy. Append  $q$  on  $[L, T]$ . Its discrepancy  $t - L - A_q(t)$  is nondecreasing and ends at  $T - L - a = E$ . Every other one-sided discrepancy is nonincreasing after  $L$ . The prefix avoids  $q$ , so appending it preserves distinctness. These arguments use absolute continuity and monotonicity, and do not require piecewise constant input rates.  $\square$

The fractional-linear recurrence has a simple coordinate. For a coefficient  $c$  in  $j$  modes define

$$\eta_j = \frac{jc - 1}{jc + 1}, \quad c = \frac{1 + \eta_j}{j(1 - \eta_j)}.$$

Positive  $c$  is equivalent to  $-1 < \eta_j < 1$ . Substitution in (59) gives

$$\eta_j = \frac{j-2}{j} \eta_{j-1}. \quad (63)$$

In particular a negative transformed seed remains negative. Replacing it by zero is unnecessary and weakens the one-sided bound.

**Theorem 9.2** (Elementary coefficient for every spare-mode budget). *For  $1 \leq k < n$ , every  $A \in \mathcal{A}_n(T)$  admits at most  $k$  distinct activation blocks with one-sided error at most  $C_{n,k}T$ , where*

$$C_{n,k} = \frac{n(n-1) + (n-k)(n-k-1)}{nk(2n-k-1)}. \quad (64)$$

*In particular  $C_{n,k} \geq 1/n$ , with equality exactly when  $k = n-1$ .*

*Proof.* For  $k = 1$ , choose a mode with mass at least  $T/n$  and use it constantly. Its error is at most  $(n-1)T/n = C_{n,1}T$ . For  $k > 1$  use Lemma 9.1 and induction at  $(n-1, k-1)$ . To solve the recurrence put  $m = n-k+1 \geq 2$ . The single-block seed has  $mc_m = m-1$  and hence  $\eta_m = (m-2)/m$ . Iterating (63), including this seed factor, gives

$$\eta_m = \prod_{j=m}^n \frac{j-2}{j} = \frac{(n-k)(n-k-1)}{n(n-1)}.$$

The identity also holds when  $m = 2$ , since both sides are zero. Its value lies in  $[0, 1)$ ; inversion gives (64). The denominator is positive because  $k \geq 1$  and  $n > k$ . The assertion about  $1/n$  follows from the sign and zeros of  $\eta_n$ . This also verifies that the maximum-mass branch alone suffices for this elementary induction, though the more general lemma will be needed for sharper seeds.  $\square$

**Corollary 9.3** (One-sided, equal-mass, and unrestricted bounds). *Write*

$$L_{n,k} = \frac{1}{n(r_n^k - 1)}. \quad (65)$$

*For  $1 \leq k < n$  one has*

$$TL_{n,k} \leq G_{n,k}^-(T) \leq TC_{n,k}, \quad (66)$$

$$T \max\{1/n, L_{n,k}\} \leq G_{n,k}^=(T) \leq TC_{n,k}, \quad (67)$$

$$T \max\{1/(k+1), L_{n,k}\} \leq F_{n,k-1}(T) \leq T \max\{1/(k+1), C_{n,k}\}, \quad (68)$$

*where  $G_{n,k}^=(T)$  denotes the worst full error over inputs with all terminal masses  $T/n$ , using at most  $k$  blocks. More generally, the schedule in Theorem 9.2 has full error at most  $TC_{n,k}$  whenever every  $m_i \leq TC_{n,k}$ .*

*Proof.* For the uniform input, even a schedule with repetitions satisfies the block-end recurrence  $t_j \leq r_n(t_{j-1} + E)$  when its one-sided error is  $E$ . Iterating gives  $T \leq nE(r_n^k - 1)$ . Distinct greedy blocks attain this lower coefficient; their full uniform error is  $T \max\{1/n, L_{n,k}\}$  by Theorem 3.1. This input belongs to all three input classes. Every positive discrepancy satisfies  $A_i(t) - W_i(t) \leq m_i$ , proving the mass-qualified and equal-mass upper bounds. Finally Theorem 4.4 proves both unrestricted bounds from the one-sided ones and the omitted-mode threshold.  $\square$

**Corollary 9.4** (An exact plateau). *For integers  $k \geq 2$  and  $k + 1 \leq n \leq k(k + 1)/2$ ,*

$$F_{n,k-1}(T) = \frac{T}{k+1}. \quad (69)$$

*This is a guaranteed plateau interval, not a claim that it is maximal.*

*Proof.* Direct subtraction and factorization give

$$C_{n,k} - \frac{1}{k+1} = \frac{2(n-k-1)(n-k(k+1)/2)}{nk(k+1)(2n-k-1)}. \quad (70)$$

The denominator is positive. The numerator is nonpositive on the stated interval, so (68) proves equality. Its lower witness is  $k+1$  successive distinct pure modes of duration  $T/(k+1)$ . For  $k = 1$  the interval in the statement is empty; separately  $C_{n,1} = (n-1)/n$ , with equality to  $1/2$  at  $n = 2$ .  $\square$

**Proposition 9.5** (The first mode-count correction). *For fixed  $k \geq 1$  and  $n \rightarrow \infty$ ,*

$$C_{n,k} = \frac{1}{k} - \frac{k+1}{2kn} + \frac{k^2-1}{4kn^2} + O_k(n^{-3}), \quad (71)$$

$$L_{n,k} = \frac{1}{k} - \frac{k+1}{2kn} + \frac{k^2-1}{12kn^2} + O_k(n^{-3}). \quad (72)$$

*Each of  $G_{n,k}^-(T)$ ,  $G_{n,k}^+(T)$ , and  $F_{n,k-1}(T)$  consequently equals*

$$\frac{T}{k} - \frac{(k+1)T}{2kn} + O_k(Tn^{-2}). \quad (73)$$

*The gap between the displayed elementary upper and uniform lower coefficients is  $(k^2 - 1)/(6kn^2) + O_k(n^{-3})$ .*

*Proof.* Put  $x = 1/n$ . Dividing numerator and denominator of (64) by  $n^2$  expresses it as

$$\frac{2 - 2(k+1)x + k(k+1)x^2}{k(2 - (k+1)x)}.$$

Expand the denominator using the geometric series through degree two. For the lower coefficient use

$$(1-x)^{-k} = 1 + kx + \frac{k(k+1)}{2}x^2 + \frac{k(k+1)(k+2)}{6}x^3 + O_k(x^4)$$

and invert  $[(1-x)^{-k} - 1]/x$ . These calculations give (71)–(72), with bounded remainders near  $x = 0$  because the resulting rational denominators do not vanish there. For large  $n$ , both coefficients tend to  $1/k > 1/(k+1)$  and  $1/n$  is smaller. The three pairs of bounds in Corollary 9.3 therefore give the stated expansions. When  $k = 1$  both coefficient formulas equal  $1 - 1/n$  exactly; their second-order terms and gap vanish.  $\square$

## 9.1 Propagating exact seeds

**Theorem 9.6** (A general seed and the four-block improvement). *Let  $m \geq 2$  and  $\ell \geq 1$  be integers. Suppose an  $m$ -mode one-sided coefficient  $c_m > 0$  is proved using at most  $\ell$  distinct blocks. For  $n \geq m$ ,  $k = \ell + n - m$ , the propagated coefficient is*

$$C_{n,k}^{[c_m]} = \frac{1 + \theta}{n(1 - \theta)}, \quad \theta = \frac{m(m-1)}{n(n-1)} \frac{mc_m - 1}{mc_m + 1}. \quad (74)$$

In particular, for  $1 \leq \ell \leq 4$  and  $n > k \geq \ell$ , set  $m = n - k + \ell$  and use the established seed  $c_m = L_{m,\ell}$ . This yields

$$C_{n,k}^{(\ell)} = \frac{1 + \theta_{n,k}^{(\ell)}}{n(1 - \theta_{n,k}^{(\ell)})}, \quad \theta_{n,k}^{(\ell)} = \frac{m(m-1)}{n(n-1)} \left[ 2 \left( \frac{m-1}{m} \right)^\ell - 1 \right]. \quad (75)$$

Write  $C_{n,k}^* = C_{n,k}^{(4)}$  for  $k \geq 4$ . Then

$$G_{n,k}^-(T) \leq TC_{n,k}^*, \quad F_{n,k-1}(T) \leq T \max\{1/(k+1), C_{n,k}^*\}. \quad (76)$$

For every  $n > k \geq 4$ ,  $C_{n,k}^* < C_{n,k}$ . The four-block seed, and thus this improvement, retains the certificate dependency of Theorem 6.1.

*Proof.* Apply Lemma 9.1  $n - m$  times. Equation (63) telescopes to the factor  $m(m-1)/[n(n-1)]$ . The seed transform lies strictly between  $-1$  and one, and multiplication by a factor in  $(0, 1]$  preserves this interval. Thus all coefficients are positive and all denominators are positive. Distinctness is preserved at every step. Substituting  $c_m = L_{m,\ell}$  gives the bracket in (75). The seed is available for  $m \geq \ell + 1$  by the established reach theorems. The heavy-mode reduction proves the full bound.

For strict comparison, the elementary construction at  $k = 4$  satisfies

$$C_{m,4} - L_{m,4} = \frac{10m^2 - 5m + 1}{2(2m-5)(2m-1)(2m^2 - 2m + 1)} > 0 \quad (m \geq 5).$$

All denominator factors are positive, and the numerator is  $5m(2m-1) + 1 > 0$ . Further,  $f'_j(c) = (j-2)/[(j-1)(1+c)^2] > 0$  for  $j \geq 3$ . The elementary coefficient is the same recurrence from its elementary four-block seed; strict improvement therefore persists at every step. This comparison is about one-sided coefficients: the maxima in the full bounds can coincide on a plateau.  $\square$

**Corollary 9.7** (Closed consequences of the seed). *For  $k \geq 4$ , equal-terminal-mass inputs have exact worst full error  $T/n$  when  $n = k + 1$  or  $n = k + 2$ . In addition,*

$$F_{16,4}(T) = T/6. \quad (77)$$

For all  $n > k \geq 4$ , a sufficient exact-plateau criterion, involving only integers, is

$$(n+k+1)(m-1)[2(m-1)^4 - m^4] \leq (n-k-1)n(n-1)m^3, \quad m = n - k + 4. \quad (78)$$

*Proof.* The fourth-power seed transform is negative exactly at the integer mode counts  $m = 5, 6$ :  $2(4/5)^4 < 1$ ,  $2(5/6)^4 < 1$ , and  $2(6/7)^4 > 1$ , while  $(1 - 1/m)^4$  increases with  $m$ . Hence  $C_{n,k}^* < 1/n$  on the two specified diagonals. The seeded schedule then has full error at most  $T/n$  for equal terminal masses. Every schedule with  $k < n$  blocks omits a mode, so the uniform input gives the matching  $T/n$  lower bound. The broader certificate-free equal-mass band in Corollary 10.2 also includes both diagonals. Retaining the negative transform supplies the stronger one-sided coefficient below  $1/n$ , even though clipping would suffice for these full-error values.

At  $(n, k) = (16, 5)$  direct evaluation gives

$$C_{16,5}^* = \frac{588449}{3544816} < \frac{1}{6} < \frac{35}{208} = C_{16,5}.$$

Six successive pure modes give the lower bound  $T/6$ . This extends the elementary four-switch plateau through one additional mode count. At  $n = 17, k = 5$  the seeded coefficient exceeds

1/6, so this bound alone certifies no further extension there. Finally  $C_{n,k}^* \leq 1/(k+1)$  is equivalent to  $(n+k+1)\theta_{n,k}^{(4)} \leq n-k-1$ . Substituting the expression for  $\theta$  and clearing the positive denominator  $n(n-1)m^3$  gives (78). Whenever it holds, the full upper bound matches the omitted-mode lower witness. It characterizes this certificate's plateau criterion, not every parameter at which the actual minimax may have that value.  $\square$

**Proposition 9.8** (Seed-dependent second-order gap). *For fixed  $k \geq \ell$ ,  $1 \leq \ell \leq 4$ , as  $n \rightarrow \infty$ ,*

$$C_{n,k}^{(\ell)} = \frac{1}{k} - \frac{k+1}{2kn} + \frac{3k^3 - 3k - 2\ell^3 + 2\ell}{12k^2n^2} + O_{k,\ell}(n^{-3}), \quad (79)$$

$$C_{n,k}^{(\ell)} - L_{n,k} = \frac{(k-\ell)(k^2 + k\ell + \ell^2 - 1)}{6k^2n^2} + O_{k,\ell}(n^{-3}). \quad (80)$$

For  $\ell = 4$  the latter coefficient is  $(k^3 - k - 60)/(6k^2)$ . For  $k = 5$  it is  $2/5$ , half the elementary gap coefficient  $4/5$ . These are bounds on the unresolved second-order error, not an exact second-order expansion of the general minimax.

*Proof.* With  $x = 1/n$  and  $d = k - \ell$ , the transformed coefficient is

$$\frac{(1-dx)(1-(d+1)x)}{1-x} \left[ 2 \left( \frac{1-(d+1)x}{1-dx} \right)^\ell - 1 \right].$$

Expanding through degree three gives

$$\theta = 1 - 2kx + k(k-1)x^2 + \left( k(k-1) - \frac{\ell^3 - \ell}{3} \right) x^3 + O_{k,\ell}(x^4).$$

Substitute into  $x(1+\theta)/(1-\theta)$  and cancel  $x$  from the denominator. Its new constant term is  $2k > 0$ , so ordinary power-series division is valid and gives (79). Subtracting (72) and using  $k^3 - k - \ell^3 + \ell = (k-\ell)(k^2 + k\ell + \ell^2 - 1)$  gives the gap. For  $k = \ell$  the seed is exactly  $L_{n,k}$ , so in that case the entire gap is zero, not merely its displayed second-order coefficient.  $\square$

## 10 Further boundary bounds and analytic predecessors

Some simpler constructions remain useful because they need fewer certificate dependencies, apply to particular input profiles, or resolve boundary regimes. The specialized heavy-mode results for two and three switches are subsumed, with their strict mass threshold unchanged, by Theorem 4.1. Their original constructive verification programs are retained in the supplement. The following light-input result has additional consequences.

**Lemma 10.1** (Distinct greedy blocks for light inputs). *Suppose  $m_i \leq E$  for all modes, where  $E > 0$ . For every  $1 \leq k < n$ , at most  $k$  distinct blocks reach*

$$\min \left\{ T, \frac{k(n-k+1)}{n-k} E \right\} \quad (81)$$

*with full error at most  $E$  on that prefix.*

*Proof.* Put  $t_0 = 0$  and  $t_j = j(n-j+1)E/(n-j)$  for  $1 \leq j \leq k$ . These times are strictly increasing: the function  $j(n-j+1)/(n-j) = j + j/(n-j)$  increases on the integers below

$n$ . Suppose  $j - 1$  distinct modes have reached  $t_{j-1} < T$ . At  $u = \min\{T, t_j\}$  their allocations sum to at most  $(j - 1)E$ . One of the  $n - j + 1$  unused modes therefore satisfies

$$A_i(u) \geq \frac{u - (j - 1)E}{n - j + 1}.$$

The identity

$$(n - j)t_j = (n - j + 1)t_{j-1} + (n - 2j + 2)E$$

and  $u \leq t_j$  imply

$$\frac{u - (j - 1)E}{n - j + 1} \geq u - t_{j-1} - E.$$

Activate this mode from  $t_{j-1}$  to  $u$ . It was previously unused, so its endpoint discrepancy is  $u - t_{j-1} - A_i(u) \leq E$ ; during the block this discrepancy is nondecreasing. Earlier selected modes receive no additional service and preserve their one-sided bound. All positive discrepancies are bounded by their terminal masses, hence by  $E$ . Stop at  $T$ , or continue the induction.  $\square$

**Corollary 10.2** (Boundary consequences and an equal-mass band). *For every  $k \geq 1$  and  $2 \leq n \leq k$ ,*

$$F_{n,k-1}(T) \leq \frac{T}{k+1}. \quad (82)$$

*For  $k + 1 \leq n \leq 2k$ , the exact value is  $F_{n,k-1}(T) = T/(k + 1)$ . Furthermore, every input with terminal masses  $T/n$  has instance optimum  $\text{OPT}_{k-1}(A) = T/n$  throughout the band*

$$1 \leq k < n, \quad (n - k)^2 \leq k, \quad G_{n,k}^-(T) = \frac{T}{n}. \quad (83)$$

*Proof.* At  $E = T/(k + 1)$  and  $n \leq k$ , some mass is at least  $T/n > E$ . Theorem 4.1 applies directly with  $s = k - 1$ . The all-light case is impossible here and is not needed; no matching lower bound is claimed for (82). For  $k < n \leq 2k$ , either the same heavy argument applies or Lemma 10.1 reaches  $T$ , since  $k(n - k + 1)/(n - k) \geq k + 1$  is equivalent to  $n \leq 2k$ . The  $k + 1$ -pure-mode witness gives equality. For  $k = 1$  this includes  $n = 2$ ; for  $k = 2$  it gives  $n = 3, 4$ . Some of these endpoints go beyond the elementary plateau interval, though Theorem 7.2 already covers them.

For equal terminal masses, instead set  $E = T/n$ . Writing  $d = n - k > 0$ , the reach condition becomes  $k(d + 1) \geq nd = (k + d)d$ , or  $k \geq d^2$ . Lemma 10.1 therefore gives full error at most  $T/n$ . Every  $k$ -block schedule omits a mode, whose terminal mass is  $T/n$ ; this proves the lower bound for every input in this class. The band includes  $n = k + 2$  for all  $k \geq 4$ ,  $n = k + 3$  for all  $k \geq 9$ , and in general all fixed deficits  $d$  once  $k \geq d^2$ . The statement does not assert a characterization outside this band.  $\square$

**Proposition 10.3** (The dimension-free limit). *For every fixed  $k \geq 1$ ,*

$$F_{n,k-1}(T) < T/k \quad \text{for every finite } n \geq 2, \quad \sup_{n \geq 2} F_{n,k-1}(T) = T/k.$$

*Proof.* If  $n \leq k$ , use (82). If  $n > k$ , then

$$\frac{1}{k} - C_{n,k} = \frac{(k + 1)(n - k)}{nk(2n - k - 1)} > 0,$$

and both entries of the full upper bound (68) are strictly below  $1/k$ . The uniform lower coefficient tends to  $1/k$ , proving the supremum assertion. The leading dimension-free

estimate also has a direct classical rounding proof: divide the horizon into  $k$  equal slots and use the network in Lemma 4.2, without its objective requiring a repetition. Integral prefix counts lie between floor and ceiling allocations. Each endpoint error is strictly smaller than a slot length, and Lemma 2.1 preserves that error inside each slot. This explains why the leading limit is already a coarse-slot rounding consequence; the finite-mode corrections require the sharper constructions above.  $\square$

The classical unrestricted CIA estimate also gives a useful explicit comparison for every  $n \geq 2$  and  $k \geq 1$ :

$$F_{n,k-1}(T) \leq \frac{2n-3}{2n-2} \frac{T}{k}. \quad (84)$$

Indeed, apply Zeile et al. [11, Corollary 1, p. 669] to the cell averages on  $k$  equal cells. Every integer choice on these cells has at most  $k-1$  switches, and cell averaging preserves its error for the original measurable input. The source's sharp unrestricted-grid constant does not imply that (84) is sharp with this hard switch budget. For fixed small  $n$  and sufficiently large  $k$ , it improves  $T/(k+1)$ ; it also makes the classical origin of the leading dimension-free estimate explicit.

**Proposition 10.4** (A certificate-free four-block upper bound). *For  $n \geq 5$ , define*

$$U_n = \frac{n^4 - 7n^3 + 21n^2 - 29n + 15}{n(2n-3)(2n^2-6n+5)}. \quad (85)$$

*Every input admits four distinct blocks with one-sided error at most  $TU_n$ , and  $F_{n,3}(T) \leq T \max\{1/5, U_n\}$ . This proves the exact plateau  $F_{n,3}(T) = T/5$  for  $5 \leq n \leq 11$  without the four-block certificates.*

*Proof.* Use the analytic three-block seed  $L_{n-1,3}$  and apply Lemma 9.1 once. Substitution gives  $f_n(L_{n-1,3}) = U_n$ . At  $n=5$  the seed is below  $1/(n-1)$ , so the minimum-mass branch supplies the one-sided conclusion there; for  $n \geq 6$  the maximum-mass branch applies. The denominator in (85) is positive for  $n \geq 5$ . The full bound follows from the heavy-mode reduction.

The numerator of  $U_n - 1/5$ , after clearing its positive denominator, is

$$p(n) = n^4 - 17n^3 + 77n^2 - 130n + 75.$$

At  $n = 5, 6, 7, 8, 9, 10, 11$  its values are respectively  $-150, -309, -492, -645, -690, -525, -24$ . At  $n = 12 + h$  it is

$$h^4 + 31h^3 + 329h^2 + 1286h + 963 > 0 \quad (h \geq 0).$$

Thus exactly the stated integer interval satisfies  $U_n \leq 1/5$ . The omitted-mode witness gives the matching lower bound throughout it.  $\square$

The stronger exact value (50) now supersedes this global upper bound, but its analytic dependency remains useful. For a quantitative comparison,

$$U_n - L_{n,4} = \frac{6n^4 - 20n^3 + 21n^2 - 7n + 1}{(2n-3)(2n-1)(2n^2-6n+5)(2n^2-2n+1)} > 0. \quad (86)$$

Positivity follows by writing the numerator at  $n = 5 + h$  as  $6h^4 + 100h^3 + 621h^2 + 1703h + 1741$ . Power-series division gives

$$U_n = \frac{1}{4} - \frac{5}{8n} + \frac{11}{16n^2} + O(n^{-3}), \quad U_n - L_{n,4} = \frac{3}{8n^2} + O(n^{-3}).$$

For  $n \geq 12$  the latter is precisely the gap above the exact full minimax coefficient; on the plateau both full bounds equal  $1/5$ .

**Lemma 10.5** (A sufficient condition using the third largest mass). *Let  $n \geq 3$ , let  $m_{(3)}$  be the third largest terminal mass, and put  $r = n/(n-1)$ . If*

$$\max_i m_i \leq E, \quad E \geq \frac{T - m_{(3)}}{1 + r + r^2}, \quad (87)$$

*then at most three distinct blocks suffice for full error at most  $E$ .*

*Proof.* Set  $L = T - m_{(3)} - E$ . If  $L \leq 0$ , any mode among the three largest masses can be used throughout: its one-sided error is at most  $T - m_{(3)} \leq E$ , and all positive discrepancies are bounded by the masses. If  $L > 0$ , then  $L < T$ , and (87) is exactly  $L \leq (r + r^2)E = B_2(n, E)$ . Lemma 5.1 supplies at most two distinct blocks on  $[0, L]$ . At least one of the three largest-mass modes is unused. Append it on  $[L, T]$ ; its final one-sided discrepancy is at most  $T - L - m_{(3)} = E$  and is nondecreasing during its block. Other one-sided discrepancies can only decrease afterwards. The mass hypothesis bounds every positive discrepancy.  $\square$

This condition can improve on an unrestricted minimax guarantee for individual inputs. For example, at  $n = 8$  take terminal masses  $1/6$  for three modes and  $1/10$  for the other five, on horizon one. Then  $E = 245/1014$  satisfies (87), is larger than all the masses, and is strictly smaller than  $L_{8,3} = 343/1352$ . Thus every temporal profile with these masses enjoys the stronger instance guarantee, although the exact unrestricted value is the latter coefficient. An earlier global two-switch bound had coefficient

$$V_n = \frac{(n-1)^3}{3n^3 - 3n^2 - 5n + 4}, \quad n \geq 4.$$

It is strictly larger than  $L_{n,3}$  because its denominator is  $n(3n^2 - 3n + 1) - (6n - 4)$ , still positive for  $n \geq 4$ . Consequently its full bound  $T \max\{1/4, V_n\}$  is superseded by Theorem 7.2; the distinct input-dependent information worth retaining is Lemma 10.5.

## 11 Exact one-switch minimax on an arbitrary grid

Grid effects need not be an additive function of the maximum cell width. They depend on the locations of particular endpoints, and uniform relaxed controls need not maximize the error even when the grid is uniform. We now determine the full outer minimax value for every grid and every  $n \geq 3$ . Throughout this section write  $\mathcal{T} = (t_0, \dots, t_N)$ , where  $0 = t_0 < \dots < t_N = T$ .

### 11.1 An explicit formula and compressed extremizers

Let  $c = \min\{j : t_j \geq T/3\}$  and define

$$H = \min \left\{ \frac{T + t_c}{4}, \frac{T - t_{c-1}}{2} \right\}. \quad (88)$$

For each  $1 \leq a \leq b \leq N$ , abbreviate  $A = t_a$ ,  $B = t_b$ ,  $P = t_{a-1}$ ,  $Q = t_{b-1}$  and put

$$L_{ab} = \max \left\{ \frac{T}{3}, \frac{(n-1)T}{n} - B, \frac{(n-1)T - A - (n-1)B}{n} \right\}, \quad (89)$$

$$U_{ab} = \min \left\{ A, \frac{(n-2)A + B}{n}, \frac{(n-1)T}{n} - P, \frac{(n-1)T - P - (n-1)Q}{n}, \frac{T - P + (n-2)A}{n} \right\}. \quad (90)$$

Call  $(a, b)$  admissible when

$$(n-2)(T - A) \leq (n-1)B, \quad L_{ab} \leq U_{ab}. \quad (91)$$

**Theorem 11.1** (Arbitrary-grid one-switch minimax). *For  $n \geq 3$  and every strictly increasing finite grid,*

$$F_{n,1}^{\mathcal{T}} = \max\{H, \max_{(a,b)} \max_{\text{admissible}} U_{ab}\}, \quad (92)$$

where an empty inner maximum is ignored. The relaxed maximizer can be chosen with at most three time phases and two types of component: either one mode and  $n - 1$  identical modes, or two identical modes and  $n - 2$  identical modes. For rational grid data, the value and this compressed maximizer require  $O(N^2)$  rational arithmetic operations and have polynomial bit complexity in the grid encoding length and  $\log n$ .

The theorem permits  $N = 1$ , zero-length phases, constant integer controls, and arbitrary measurable relaxed inputs through Lemma 2.1. The compressed representation lists each component type once and records its multiplicity; writing all  $n$  component functions takes at least linear time in  $n$ .

*Proof.* We first derive a finite family of closed linear programs. Order the terminal masses  $m_1 \geq \dots \geq m_n$ . For fixed initial mode  $p$  and switching time  $\tau$ , a largest-mass mode other than  $p$  is an optimal final mode. Indeed, in (15), this replacement reduces the final deficit, leaves the initial deficit unchanged, and cannot increase the largest omitted mass. Thus only  $p \rightarrow 1$  with  $p \neq 1$  and  $1 \rightarrow 2$  need be considered.

For  $1 \leq a \leq b \leq N$ , use variables  $u_j, v_j, m_j$  ( $j = 1, 2, 3$ ) and  $E$ , where component 3 is repeated  $n - 2$  times. With weights  $w = (1, 1, n - 2)$ , impose

$$0 \leq u_j \leq v_j \leq m_j \quad (j = 1, 2, 3), \quad m_1 \geq m_2 \geq m_3, \quad E \geq T/3, \quad (93a)$$

$$\sum_j w_j u_j = t_a, \quad \sum_j w_j v_j = t_b, \quad \sum_j w_j m_j = T, \quad (93b)$$

$$T - t_a \leq E + m_1 \leq T - t_{a-1}, \quad T - t_b \leq E + m_2 \leq T - t_{b-1}, \quad (93c)$$

$$u_2 + E \leq t_a, \quad v_1 + E \leq t_b. \quad (93d)$$

The *all-initial-modes family* also imposes  $u_3 + E \leq t_a$ . The *two-large-modes family* instead imposes  $m_2 \geq E$ . Both maximize  $E$ ; an infeasible program is ignored. In particular, the first family does not impose  $m_2 \leq E$ .

Every feasible point gives a valid relaxed cumulative control by linear interpolation through  $0, t_a, t_b, T$ . Nonnegative increments and the sum constraints give nonnegative rates summing to one. Equal times force equal states, so empty phases can be removed. For  $p \rightarrow 1$ , a grid switch before  $t_a$  has final deficit  $T - m_1 - \tau \geq T - m_1 - t_{a-1} \geq E$ . A switch at or after  $t_a$  has initial deficit at least  $t_a - u_p$ , since  $t - A_p(t)$  is nondecreasing. The same argument for  $1 \rightarrow 2$  uses  $t_b, v_1$ . In the first family this obstructs every schedule by dominance. In the second family any schedule omitting mode 1 or 2 has error at least  $m_2 \geq E$ , and the two remaining orders are obstructed by the common constraints. Constant schedules are included. Hence each feasible point is a lower-bound witness of value  $E$ .

Conversely, let  $F = F_{n,1}^{\mathcal{T}}$ , attained by compactness. Giving three modes mass  $T/3$  proves  $F \geq T/3$ . Suppose  $F > T/3$  and fix  $T/3 < E < F$ . For  $j = 1, 2$ , let  $k_j$  be the first grid index satisfying  $t_{k_j} \geq T - E - m_j$ . Both indices are positive:  $k_1 = 0$  would give the constant mode-1 schedule error  $T - m_1 \leq E$ . Set  $a = k_1 \leq k_2 = b$ . There are at most two masses exceeding  $E$ . If  $m_2 \leq E$ , every  $p \rightarrow 1$  and  $1 \rightarrow 2$  omits only masses at most  $E$ . At its first eligible grid time its final deficit is at most  $E$ , so its initial deficit must exceed  $E$ . If  $m_2 > E$ , apply this reasoning only to  $2 \rightarrow 1$  and  $1 \rightarrow 2$ ; their omitted masses are at most  $E$  since  $3E > T$ . Averaging modes  $3, \dots, n$  preserves all required inequalities and yields a feasible point in the respective closed program. Let  $E \uparrow F$ , choose a fixed family and pair  $(a, b)$  along a subsequence, and take a convergent subsequence of the bounded endpoint data. This gives

a point with value  $F$ . If  $F = T/3$ , take  $E = T/3$ ,  $a = b = c$ ,  $m_1 = m_2 = E$ ,  $m_3 = E/(n-2)$ ,  $u_1 = v_1 = u_2 = v_2 = (t_c - E)/2$ , and  $u_3 = v_3 = E/(n-2)$ . Since  $E \leq t_c \leq 3E$ , this is feasible in the second family. The maximum of the two program families is therefore  $F$ .

We next eliminate the two-large-modes family. At  $B = t_b$ , both selected modes have allocation at most  $B - E$ ; for mode 2 this follows by monotonicity of its deficit after  $t_a$ . All remaining modes have total mass at most  $T - 2E$ , giving  $B \leq 2(B - E) + T - 2E$  and  $E \leq (T + B)/4$ . The cutoff and  $m_2 \geq E$  also give  $E \leq (T - Q)/2$ . If  $B < T/3$  or  $Q > T/3$ , one of these bounds is below  $T/3$ . Thus only  $b = c$  can give more than  $T/3$ , and the entire family is bounded by  $H$ . For attainment let  $C = t_c$ ,  $E = H$ , and assign two identical modes total mass  $E$  each. At  $C$ , give each of them

$$x = \max\{0, (C - T + 2E)/2\}; \quad (94)$$

each of the other  $n - 2$  modes receives cumulative mass  $\min(C, T - 2E)/(n - 2)$  and terminal mass  $(T - 2E)/(n - 2)$ . Interpolate on  $[0, C]$  and  $[C, T]$ . The inequalities  $E \geq T/3$ ,  $E \leq C$ ,  $4E \leq T + C$ , and  $T - C \leq 2E \leq T - t_{c-1}$  verify every constraint with  $a = b = c$ . This proves that the second family contributes exactly  $H$ .

For the remaining contribution only one distinguished mode is needed. Return to the coverage argument with  $m_2 \leq E < F$  and average all modes other than mode 1. Each averaged initial deficit still exceeds  $E$  at  $t_a$ . Their common terminal mass is at most the old  $m_2$ , so the first eligible time for  $1 \rightarrow p$  moves weakly later. The initial deficit of mode 1 is nondecreasing and remains above  $E$  there. Thus this symmetrization preserves the obstruction, with a possibly new  $b \geq a$ . Limits again cover boundary values; if the overall maximum comes only from the other family,  $H$  already covers it.

Write  $M = m_1$ ,  $x = A_1(A)$ ,  $y = A_1(B)$ . Every other mode has values  $(A - x)/(n - 1)$ ,  $(B - y)/(n - 1)$ ,  $(T - M)/(n - 1)$ . The failure constraints are  $x \geq (n - 1)E - (n - 2)A$  and  $y \leq B - E$ . Monotonicity is equivalent to

$$0 \leq x \leq y \leq M, \quad 0 \leq A - x \leq B - y \leq T - M.$$

Together with the cutoff constraints and  $M \geq T/n$ , such states exist if and only if  $E \leq A$ ,  $nE \leq (n - 2)A + B$ , and

$$\begin{aligned} \max\{T/n, T - A - E, (n - 1)(E + Q) - (n - 2)T, (n - 1)E - (n - 2)A\} \\ \leq M \leq \min\{T - P - E, (n - 1)(E + B) - (n - 2)T\}. \end{aligned} \quad (95)$$

Necessity follows from the displayed monotonicity and cutoff inequalities. For sufficiency choose any  $M$  in this interval and set

$$x = \max\{0, (n - 1)E - (n - 2)A, A - T + M\}, \quad y = \max\{x, B - T + M\}. \quad (96)$$

The assumptions give  $x \leq \min(A, M, B - E)$  and  $y \leq \min(M, B - E)$ . They also give  $y - x \leq B - A$  and  $B - y \leq T - M$ , proving the required monotonicity. For example,  $B - T + M \leq B - E$  follows from  $M \leq T - P - E \leq T - E$ ;  $x \leq B - E$  follows from  $nE \leq (n - 2)A + B$  and this same upper bound. These observations also cover  $a = b$  and  $b = N$ .

Compare the four lower endpoints of (95) with its two upper endpoints. The comparisons involving  $T - A - E$  versus  $T - P - E$ , and  $(n - 1)(E + Q) - (n - 2)T$  versus  $(n - 1)(E + B) - (n - 2)T$ , hold automatically. The comparison of  $(n - 1)E - (n - 2)A$  with the second upper endpoint is exactly the first inequality of (91). The remaining five comparisons, together with  $E \geq T/3$ ,  $E \leq A$  and  $nE \leq (n - 2)A + B$ , are exactly  $L_{ab} \leq E \leq U_{ab}$ . Maximization proves (92); (95)–(96) reconstruct its first-family extremizer with at most three phases. Each pair uses a fixed number of rational operations on the original grid endpoints and  $n$ . The formulas have fixed algebraic depth, so intermediate bit lengths are polynomial in the input length. Comparing  $O(N^2)$  candidates proves the complexity claim.  $\square$

## 11.2 Unit-grid consequences and the need for both families

**Corollary 11.2** (Three modes, one switch). *For  $N \geq 2$  unit cells,*

$$F_{3,1}^{\text{unit}}(N) = \begin{cases} k, & N = 3k, \\ k + \frac{1}{2}, & N = 3k + 1 \quad (k \geq 1), \\ k + \frac{3}{4}, & N = 3k + 2 \quad (k \geq 0). \end{cases} \quad (97)$$

*For  $N = 1$  the value is  $2/3$ .*

*Proof.* The one-cell case is Proposition 2.3. For  $N = 3k$ , choosing the second-largest mass first until  $k$  and the largest mass last gives all three error terms at most  $k$ ; three equal masses give the matching omitted-mode lower bound. For the other cases put  $L = k + 1$  and let  $E$  be the claimed value. Direct substitution gives

$$\begin{aligned} E \geq N/3, \quad N - L \leq 2E, \quad N + L \leq 4E, \quad E + L \geq 2N/3, \\ 3E \geq 2L, \quad 3E + k + 2L \geq 2N. \end{aligned} \quad (98)$$

There are at most two masses greater than  $E$ . If there are two, their allocations at  $L$  sum to more than  $L - N + 2E \geq 2(L - E)$ . One of them, used first, therefore has initial deficit at most  $E$  at  $L$ ; the other, used last, has final deficit less than  $N - E - L \leq E$ . The omitted mass is at most  $E$ . Otherwise let  $q, r$  have the largest and second-largest masses. If  $m_q \geq N - E - k$ , the order  $r \rightarrow q$  at  $k$  works. If not, then  $m_q < N - E - k$ , while  $m_q \geq N/3$  ensures that any  $p \rightarrow q$  at  $L$  has final deficit at most  $E$ . Such a schedule works if some  $p \neq q$  has  $A_p(L) \geq L - E$ . In the remaining case  $A_q(L) > 2E - L \geq L - E$ , and

$$m_r \geq (N - m_q)/2 > (E + k)/2 \geq N - E - L.$$

Thus  $q \rightarrow r$  at  $L$  works. These schedules omit only masses at most  $E$ , establishing the upper bound.

For  $N = 3k + 1$ , use  $k$  pure cells of mode 3, one cell with rates  $(1/2, 1/2, 0)$ , then  $k$  pure cells of mode 1 and  $k$  of mode 2. Both selected large masses equal  $E = k + 1/2$ , with allocation  $1/2 = L - E$  at  $L$ . Omitting either gives error  $E$ ; selecting both gives final deficit at least  $E$  for switches at  $\tau \leq k$ , and initial deficit at least  $E$  for  $\tau \geq L$ . For  $N = 3k + 2$ , replace the mixed cell by  $(1/4, 1/4, 1/2)$  and append one final cell  $(1/2, 1/2, 0)$ . The two large masses are now  $E = k + 3/4$ , and each has allocation  $1/4 = L - E$  at  $L$ . The same argument applies: early switches have final deficit at least  $k + 5/4 > E$ , and late switches have initial deficit at least  $E$ . Empty pure blocks when  $k = 0$  cause no change in the argument.  $\square$

The corrections to  $N/3$  are respectively 0,  $1/6$ , and  $1/12$ ; they are not a fixed half-cell correction.

**Example 11.3** (Five modes and nine unit cells). The exact value is  $F_{5,1}^{\text{unit}}(9) = 17/5$ . One extremizer has rates

| segment  | mode 1 | each of modes 2, 3, 4, 5 |
|----------|--------|--------------------------|
| $[0, 4]$ | $2/5$  | $3/20$                   |
| $[4, 5]$ | 0      | $1/4$                    |
| $[5, 9]$ | $1/4$  | $3/16$                   |

Its terminal masses are  $(13/5, 8/5, 8/5, 8/5, 8/5)$ . Every  $p \rightarrow 1$  with  $p \neq 1$  has final deficit at least  $17/5$  when  $\tau \leq 3$ , and initial deficit at least  $4 - 3/5 = 17/5$  when  $\tau \geq 4$ . For  $1 \rightarrow 2$ , the

analogous cutoff is between 4 and 5: the final deficit at 4 is  $17/5$ , and the initial deficit at 5 is  $5 - 8/5 = 17/5$ . Dominance proves the lower bound. A short independent upper proof is useful. Sort the masses. If  $m_1 \geq 13/5$ , use  $2 \rightarrow 1$  at time 3: the initial deficit is at most 3, the omitted mass is at most  $m_3 \leq 3$ , and the final deficit is  $6 - m_1 \leq 17/5$ . Otherwise  $m_2 \geq (9 - m_1)/4 > 8/5$ . At time 4 choose  $p$  with  $A_p(4) \geq 4/5$  and finish in a largest-mass mode  $q \neq p$ . The initial deficit is at most  $16/5$ , the final deficit is  $5 - m_q < 17/5$ , and omitted masses are below  $13/5$ . This exceeds both the uniform-input value  $16/5$  and the three-mode worst-case value 3, disproving the proposed reduction to those two families of inputs.

The admissibility conditions in Theorem 11.1 also matter on nonuniform grids. For

$$n = 9, \quad \mathcal{T} = (0, 19/2, 61/6, 265/24, 481/24, 2669/120),$$

a proposed first-family simplification retaining only the upper bounds  $((n-2)A + B)/n$  and  $((n-1)T - P - (n-1)Q)/n$  gives  $2077/216$ , whereas the exact first-family maximum is  $2593/270$ . This discrepancy explains why the other cutoff and feasibility conditions cannot be discarded; the complete formula gives an exact answer without an LP solver. The bundled rational checker verifies these values, reconstructs extremizers, and checks their original absolute discrepancies by exhaustive schedule enumeration. Independently assembled LP and MILP comparisons are supplementary numerical checks, not prerequisites for the analytic theorem.

## 12 Three-mode floor chambers and exact small-grid values

For three modes, strict cumulative-floor chambers have a particularly small combinatorial description. It determines the exact minimax values on five, six, and seven equal cells at the budgets considered below, closes a continuous boundary case, and identifies a concrete obstruction to extending the five-cell construction to arbitrary budgets.

### 12.1 A complete three-state description

Work on  $N$  unit cells and suppose initially that  $A_i(j) \notin \mathbb{Z}$  for every mode  $i \in \{0, 1, 2\}$  and prefix  $1 \leq j \leq N$ . Put  $f_j = (\lfloor A_i(j) \rfloor)_i$  and  $\sigma_j = j - \sum_i f_{j,i}$ . Since all three fractional parts are strictly between zero and one,  $\sigma_j \in \{1, 2\}$ . An integer word has discrepancy less than one exactly when its prefix counts lie coordinatewise between  $f_j$  and  $f_j + \mathbf{1}$ . Its possible count vectors at this prefix are therefore exactly the three vectors

$$f_j + o_{\sigma_j}(u), \quad u \in \{0, 1, 2\}, \quad o_1(u) = e_u, \quad o_2(u) = \mathbf{1} - e_u. \quad (99)$$

The index  $u$  labels a count vector; it need not be the active mode.

**Proposition 12.1** (Realizable floor histories). *A sequence of integer floors is realized by a relaxed control with all prefix coordinates nonintegral if and only if it starts with  $f_1 = 0$ ,  $\sigma_1 = 1$  and, at each subsequent cell, its increment  $d = f_j - f_{j-1}$  has one of the following forms:*

| $\sigma_{j-1}$ | $\sigma_j$ | $d$                                    |
|----------------|------------|----------------------------------------|
| 1              | 1          | $e_p \quad (p = 0, 1, 2)$              |
| 1              | 2          | 0                                      |
| 2              | 1          | $\mathbf{1} - e_p \quad (p = 0, 1, 2)$ |
| 2              | 2          | $e_p \quad (p = 0, 1, 2).$             |

(100)

For each such transition there is an edge from count node  $u$  to count node  $v$ , labeled by active mode  $a$ , precisely when

$$d + o_{\sigma_j}(v) - o_{\sigma_{j-1}}(u) = e_a. \quad (101)$$

The full integer words in this layered graph are exactly the words with discrepancy less than one throughout this floor chamber.

*Proof.* Every cell allocation lies in  $[0, 1]$ , so each floor increment is zero or one. Its sum is  $1 + \sigma_{j-1} - \sigma_j$ , giving precisely (100); the initial floors vanish since three positive nonintegral allocations sum to one at the first prefix. An integer word increments its count vector by one unit vector per cell, which proves (101) and the word correspondence whenever the chamber exists.

It remains to prove that every displayed history really exists. Each of its transition graphs has an incoming and an outgoing edge at every node. This follows directly from (101). More explicitly, for  $1 \rightarrow 1$  with increment  $e_p$ , all self edges have label  $p$  and  $u = p$  can move to any  $v$  with label  $v$ . For  $1 \rightarrow 2$ , the edges are  $u \neq v$ , labeled by the third mode. For  $2 \rightarrow 1$  with increment  $1 - e_p$ , the edges are  $u = p$ , labeled  $v$ , or  $v = p$ , labeled  $u$ . For  $2 \rightarrow 2$  with increment  $e_p$ , all self edges have label  $p$  and any  $u$  can move to  $v = p$  with label  $u$ . All three initial nodes are available, with first active mode  $u$ . Induction forward and backward shows that every node at every layer lies on a full path. Average the cell allocations of all these finitely many paths. At every prefix and coordinate, the three count nodes include both the floor and ceiling, and all three occur on full paths. Their average is therefore strictly between these endpoints. This average is a valid simplex-valued relaxed control realizing the prescribed floors.  $\square$

This proof gives a realizability criterion without a linear program or a search through infeasible floor arrays. The numbers of histories ending in states 1, 2 are the row vector

$$(1, 0) \begin{pmatrix} 3 & 1 \\ 3 & 3 \end{pmatrix}^{N-1}. \quad (102)$$

The total counts for  $N = 1, \dots, 7$  are 1, 4, 18, 84, 396, 1872, 8856.

The least switch count in one chamber is also computed by a small exact recursion. Let  $C_j(u, a)$  be the least count among its prefix paths ending at node  $u$  with last active mode  $a$ , and use  $+\infty$  for an empty set. Initialize  $C_1(u, a) = 0$  if  $a = u$  and  $+\infty$  otherwise. For each edge  $u \rightarrow v$  labeled  $b$ , update

$$C_j(v, b) = \min_{u, a: u \rightarrow v \text{ has label } b} \{C_{j-1}(u, a) + \mathbf{1}_{a \neq b}\}. \quad (103)$$

Every prefix word has exactly one predecessor, so induction proves this recursion, including the unrestricted initial activation convention. The answer is  $\min_{u, a} C_N(u, a)$ .

## 12.2 The exact five-cell theorem and a continuous consequence

**Theorem 12.2** (Five equal cells). *For three modes and five cells of common length  $\Delta > 0$ ,*

$$F_{3,2}^T = \Delta. \quad (104)$$

*The same value holds when the relaxed input is any measurable control.*

*Proof.* Scale to unit cells. Enumerate the four transition types in (100) for four successive steps from  $(f_1, \sigma_1) = (0, 1)$ , taking all three choices of  $p$  where present. For every resulting history apply (101) and (103). The exact integer computation gives

| minimum switches          | 0 | 1   | 2   | total |
|---------------------------|---|-----|-----|-------|
| number of strict chambers | 3 | 138 | 255 | 396   |

There are no other histories by Proposition 12.1, and all transition branches are enumerated. The standard-library checker `verification/stage04/check_floor_chambers.py` implements these three displayed formulas and verifies each terminal minimum is at most two. Thus this is a finite integer certificate, not a numerical estimate. An independent bundled checker also enumerates all  $3^5 = 243$  words against all  $1 \cdot 4 \cdot 9 \cdot 16 \cdot 25 = 14400$  candidate floor arrays and recovers the same chambers and switch counts. For that alternative enumeration, strict feasibility is equivalent to each bounded prefix coordinate attaining both its floor and ceiling among the feasible integer words. Necessity follows by decomposing a feasible fractional prefix flow into integral flows; sufficiency follows by averaging all the feasible words. This gives an independent coverage check rather than relying only on agreement of the final count.

Every control with nonintegral prefixes consequently has a two-switch word with error strictly below one. To cover integral boundary coordinates, perturb the cell rates toward  $(1/7, 2/7, 4/7)$ . Every coordinate of this constant control is nonintegral at prefixes  $1, \dots, 5$ . Along a sequence of positive perturbations tending to zero all prefix coordinates avoid integers: each coordinate has only finitely many forbidden perturbation parameters unless constant, in which case it is already nonintegral. The minimum discrepancy over the finite set of two-switch words is continuous, so the original control has error at most one. For arbitrary  $N$  the same argument uses, for example,  $(1/p, 2/p, (p-3)/p)$  with a prime  $p > \max\{N, 3\}$ .

For sharpness, take the pure relaxed word 01010. Its integer prefix counts force any word with error less than one to have the same prefix counts and hence the identical five letters, requiring four switches. Thus every two-switch word has error at least one. Cell averaging and scaling complete the proof.  $\square$

**Corollary 12.3** (Three modes, two continuous switches). *For every  $T > 0$ ,*

$$F_{3,2}(T) = T/5. \quad (105)$$

*Proof.* The upper bound follows from (8) and Theorem 12.2 on five cells of length  $T/5$ . The lower bound is the case  $(n, s) = (3, 2)$  of Sager and Zeile [9, Proposition 4, p. 612, equation (7.4)]. Here is a direct alternative witness. Scale to  $T = 5$  and take five pure unit phases 01201, with terminal masses  $(2, 2, 1)$ . If a schedule had error  $E < 1$ , it would need to use all three modes, so with at most three blocks each mode would occur exactly once. Let  $i$  be its final mode, starting at  $v$ . The first times when the three relaxed allocations reach one are  $d = (1, 2, 3)$ . Since mode  $i$  is unserved before  $v$ , necessarily  $v < d_i$ . On the other hand its final signed error gives  $v \geq 5 - m_i - E > 5 - m_i - 1$ , whose values are  $(2, 2, 3)$  and are coordinatewise at least  $d$ . This is a contradiction. The order 012 with switches at 2, 4 has error exactly one, as endpoint monotonicity or direct integration verifies.  $\square$

This adds the boundary  $n = 3$  to the continuous two-switch result of Theorem 7.2. It concerns full absolute error and does not assert uniform extremality for the one-sided three-block problem.

**Corollary 12.4** (Six equal cells). *For three modes, six cells of length  $\Delta$ , and at most three switches,  $F_{3,3}^T = \Delta$ .*

*Proof.* The same complete transition enumeration at depth six gives 3, 255, 1377, 237 chambers with respective minimum switch counts 0, 1, 2, 3, totaling 1872. Thus every strict chamber admits a three-switch word of error less than one after scaling to  $\Delta = 1$ . The preceding perturbation argument covers its boundary. The pure relaxed word 010101 gives the reverse bound: error below one would force all five switches of that word.  $\square$

### 12.3 The sharp seven-cell value

**Proposition 12.5** (A three-switch grid instance). *Let  $u = (1/3, 1/3, 1/3)$  and take seven unit-cell rate vectors*

$$(u, e_0, e_1, e_2, e_0, u, e_2). \quad (106)$$

*The optimal grid discrepancy with at most three switches is  $4/3$ . Consequently  $F_{3,3}^{\text{unit}}(7) \geq 4/3$ .*

*Proof.* The cumulative vectors, multiplied by three, are

$$(1, 1, 1), (4, 1, 1), (4, 4, 1), (4, 4, 4), (7, 4, 4), (8, 5, 5), (8, 5, 8).$$

Each coordinate has fractional part  $1/3$  or  $2/3$ . An integer outside its floor–ceiling pair therefore differs from it by at least  $4/3$ . Any word of discrepancy less than  $4/3$  must lie in this floor chamber. For clarity, the entire recursion (103) for this chamber is displayed below. Each parenthesized triple is one row of the matrix  $C_j(u, a)$ , with rows indexed by  $u$  and columns by  $a$  in the order 0, 1, 2:

| $j$ | row 0                 | row 1                 | row 2                 |
|-----|-----------------------|-----------------------|-----------------------|
| 1   | $(0, \infty, \infty)$ | $(\infty, 0, \infty)$ | $(\infty, \infty, 0)$ |
| 2   | $(0, \infty, \infty)$ | $(1, 1, \infty)$      | $(1, \infty, 1)$      |
| 3   | $(1, 1, \infty)$      | $(\infty, 1, \infty)$ | $(\infty, 2, 2)$      |
| 4   | $(3, \infty, 2)$      | $(\infty, 2, 2)$      | $(\infty, \infty, 2)$ |
| 5   | $(3, \infty, \infty)$ | $(3, 3, \infty)$      | $(3, \infty, 2)$      |
| 6   | $(\infty, 3, 4)$      | $(3, \infty, 4)$      | $(3, 4, \infty)$      |
| 7   | $(\infty, \infty, 4)$ | $(\infty, \infty, 4)$ | $(4, 4, 4)$           |

The final minimum is four, proving the lower bound under three switches. The word 0011022 has three switches and direct evaluation of its integer prefix counts gives maximum discrepancy  $4/3$ , proving attainment. The checker independently enumerates all  $3^7 = 2187$  words: exactly 104 belong to the floor chamber and their minimum switch count is four.  $\square$

The exceptional chamber in this example is sufficient to determine the full seven-cell minimax value as well.

**Theorem 12.6** (Seven equal cells). *For three modes, seven cells of length  $\Delta$ , and at most three switches,*

$$F_{3,3}^{\mathcal{T}} = 4\Delta/3. \quad (107)$$

*Proof.* Scale to  $\Delta = 1$ . Exhaustive evaluation of (100)–(103) gives

| minimum switches          | 0 | 1   | 2    | 3    | 4 | total |
|---------------------------|---|-----|------|------|---|-------|
| number of strict chambers | 3 | 414 | 4542 | 3891 | 6 | 8856  |

The six chambers requiring four switches are exactly the six mode permutations of the floor history in Proposition 12.5:

$$(0, 0, 0), (1, 0, 0), (1, 1, 0), (1, 1, 1), (2, 1, 1), (2, 1, 1), (2, 1, 2). \quad (108)$$

The checker tests equality of these two sets of histories, rather than only their cardinalities. Every other strict chamber has a three-switch word of discrepancy less than one.

Consider (108); relabeling handles its permutations. The following words have at most three switches:

$$w^0 = 1200022, \quad w^1 = 0021122, \quad w^2 = 0011220.$$

The prefix counts of  $w^i$  belong to every required floor–ceiling pair except coordinate  $i$  at time  $i + 2$ , where their count is zero instead of one or two. This is a direct integer check over the displayed floor history. Consequently, for any relaxed control in this chamber,

$$D(A, w^0) \leq \max\{1, A_0(2)\}, \quad D(A, w^1) \leq \max\{1, A_1(3)\}, \quad D(A, w^2) \leq \max\{1, A_2(4)\}.$$

Monotonicity and conservation give  $A_0(2) + A_1(3) + A_2(4) \leq \sum_i A_i(4) = 4$ . At least one of these three coordinates is therefore at most  $4/3$ , and its corresponding word proves the desired upper bound. Generic perturbation and continuity cover all boundary profiles. Proposition 12.5 supplies the matching lower bound, and cell averaging includes measurable inputs.  $\square$

Thus error one on  $2s + 1$  unit cells fails already at  $s = 3$ , while the sharp replacement there is  $4/3$ . At these parameters, this value satisfies the proposed upper bound  $3/2$  but refutes the equality asserted in Conjecture 1 of Sager and Zeile [9, equation (7.6), p. 615]. The continuous consequences are

$$\frac{T}{7} \leq F_{3,3}(T) \leq \frac{T}{6}. \quad (109)$$

The upper bound follows from Corollary 12.4 and (8). The lower bound is Sager and Zeile [9, Proposition 4, p. 612]; it can also be checked directly on that source’s pure seven-unit-cell word 0120011. If  $E < 1$ , modes 0, 1, 2 must occur before times 1, 2, 3, respectively. Mode 0 must also occur in (3, 5), and mode 1 in (5, 7), since otherwise the corresponding error changes by two across that interval. A single activation of mode 0 spanning its early and late occurrences would cover [1, 3], where its relaxed rate is zero, again changing its error by two. A single activation of mode 1 would cover [2, 5], changing its error by three. Both changes exceed  $2E$ . Thus modes 0 and 1 each need two activations and mode 2 needs one, requiring at least five blocks. Scaling proves the lower bound. The exact continuous three-switch value remains undetermined; transferring the seven-cell bound gives only  $4T/21$ .

## 12.4 The scope of the published lower-bound correction

Sager and Zeile [9, Corollary 5, p. 611] state, for  $n_\omega > 2$  and  $1 \leq \sigma_{\max} \leq N - 2$ , the lower bound

$$\theta^{\max} \geq \frac{N + \sigma_{\max} + 1}{3 + 2\sigma_{\max}} \bar{\Delta}. \quad (110)$$

At  $(n_\omega, N, \sigma_{\max}) = (3, 5, 2)$  this is  $8\Delta/7$ , whereas Theorem 12.2 gives exactly  $\Delta$ . All the stated parameter restrictions hold. Even if the source’s outer maximum is read as varying the grid with fixed horizon  $5\Delta$ , five cells, and maximum width  $\Delta$ , every cell must equal  $\Delta$ . Thus that reading does not remove the contradiction.

The preceding attainment argument in Corollary 4, p. 610, uses  $N = j(3 + 2\sigma_{\max}) + \sigma_{\max} + 2$  with integer  $j \geq 0$ . It does not establish attainment at every  $N$ ; this restriction is absent from Corollary 5. The five-cell theorem disproves (110) as written. This correction is separate from the counterexamples to Conjecture 1; it supplies no replacement lower formula for the full parameter range.

## 13 Exact algorithms for a supplied input

The minimax results quantify the hardest input. We now optimize the schedule for a supplied input on an arbitrary grid  $\mathcal{T} = (x_0, \dots, x_N)$ . The data are rational integrated cell allocations  $a_{ji} = A_i(x_j) - A_i(x_{j-1})$ , with  $a_{ji} \geq 0$  and  $\sum_i a_{ji} = \Delta_j$ . By Lemma 2.1, these

data determine the error of every grid schedule even when the original relaxed control varies inside a cell. Arithmetic-operation bounds below count rational operations; the rational bit lengths are also polynomial in the input length for each fixed switch budget.

### 13.1 One switch and an optimal final block

Order the modes by nonincreasing terminal mass, breaking ties consistently, and denote the first two by  $h_1, h_2$ . Only these leaders, and a third leader for evaluating omitted maxima, need be selected; a full sort is unnecessary.

**Theorem 13.1** (Three candidates per boundary). *For  $n \geq 3$  and a fixed internal boundary  $t$ , an optimal ordered pair of distinct initial and final modes is among*

$$(h_1, h_2), \quad (h_2, h_1), \quad (p_t, h_1), \quad p_t \in \operatorname{argmax}_{i \notin \{h_1, h_2\}} A_i(t).$$

*For  $n = 2$  the first two pairs suffice. Comparing these candidates at all permitted internal boundaries with the best constant schedule solves the at-most-one-switch problem in  $O(nN)$  arithmetic operations. The same bound holds with a prescribed initial mode and any subset of permitted boundaries.*

*Proof.* By (15), the error for  $p \rightarrow q$  at  $t$  is

$$\max\left\{\max_{i \notin \{p, q\}} m_i, t - A_p(t), T - t - m_q\right\}.$$

Replacing  $q$  by a largest-mass mode other than  $p$  cannot increase any term: the newly omitted mode has no larger mass than the mode removed from the omitted set. Thus the final mode can be  $h_2$  if  $p = h_1$  and  $h_1$  otherwise. The first two initial choices give the first two pairs. For  $p \notin \{h_1, h_2\}$  the omitted maximum is  $m_{h_2}$  and the final-mode term is fixed; maximizing  $A_p(t)$  gives the third pair.

Select at most three mass leaders in  $O(n)$  time and form the cumulative data in  $O(nN)$  operations. Each permitted boundary needs one scan to find  $p_t$  and then a constant number of error evaluations. A largest-mass mode is the best constant choice, with error  $T - m_{h_1}$ . If the initial mode is fixed to  $p$ , compare only its constant schedule and  $p \rightarrow q_p$ , where  $q_p$  has largest mass among the other modes. With no eligible boundaries, only the constant comparison remains.  $\square$

For a fixed initial mode  $p$  and its final choice  $q_p$ , the two varying terms are  $L_p(t) = t - A_p(t)$  and  $R_p(t) = T - t - m_{q_p}$ . Their difference  $2t - A_p(t) - T + m_{q_p}$  is strictly increasing, since its increase between  $u < v$  is at least  $v - u$ . Before its unique crossing, if one exists,  $\max(L_p, R_p)$  decreases; after it, that maximum is nondecreasing. The omitted-mass term is constant. Consequently the two permitted boundaries adjacent to the crossing suffice, with the first or last boundary when the crossing lies outside the permitted range. Binary search over a sorted boundary list gives  $O(n \log(N + 1))$  additional queries once the cumulative data and mass leaders are available for random access. This bound excludes reading the raw  $nN$  input values and preparing the boundary list.

Permitted boundaries can enforce a common minimum duration  $d$  for both nonempty runs by retaining  $d \leq t \leq T - d$ , provided constants are allowed and  $T \geq d$ . The dominance theorem does not by itself permit arbitrary mode-specific dwell requirements or restrictions coupling the two labels. The general algorithm below handles mode-specific minimum dwell times.

**Lemma 13.2** (Best completion of a fixed prefix). *Fix a schedule on  $[0, u]$ , where  $u < T$ , with discrepancy  $E_0$  and service  $c_i = W_i(u)$ . Put  $r_i = m_i - c_i$ , so that  $\sum_i r_i = T - u$ . Completing it in mode  $j$  has exact error*

$$\max\{E_0, \max_{i \neq j} r_i, T - u - r_j\}. \quad (111)$$

*Among any fixed nonempty set of eligible final modes, a mode with largest residual  $r_j$  is optimal. Residuals may be negative.*

*Proof.* Every inactive discrepancy increases to  $r_i$ , while the active discrepancy decreases to  $r_j - (T - u)$ . The other endpoints are already bounded by  $E_0$ , which proves the formula. Replacing  $j$  by an eligible mode with larger residual decreases the last term and cannot increase the omitted residual maximum. This argument is unaffected by negative residuals.  $\square$

This lemma optimizes the last block after a fixed prefix; it supplies no greedy rule for choosing that prefix. The eligible set is held fixed during this comparison.

### 13.2 Assigning modes to a fixed set of blocks

Fix  $k$  positive consecutive blocks with grid endpoint indices  $0 = b_0 < b_1 < \dots < b_k = N$ , and lengths  $\ell_j = x_{b_j} - x_{b_{j-1}}$ . For mode  $i$  and a subset  $U \subseteq [k]$ , define

$$c_i(U) = \max_{1 \leq j \leq k} \left| A_i(x_{b_j}) - \sum_{\substack{h \in U \\ h \leq j}} \ell_h \right|. \quad (112)$$

This is exactly the component error when mode  $i$  occupies those blocks, including disconnected subsets. In particular  $c_i(\emptyset) = m_i$ : unused modes still contribute to the objective.

**Proposition 13.3** (Subset assignment). *The optimum for these fixed block boundaries is obtained from*

$$D_0(\emptyset) = 0, \quad D_0(S) = +\infty \ (S \neq \emptyset), \quad D_i(S) = \min_{U \subseteq S} \max\{D_{i-1}(S \setminus U), c_i(U)\}, \quad (113)$$

*as  $D_n([k])$ . After cumulative preprocessing, its arithmetic cost is  $O(n(k2^k + 3^k))$  and its storage, including traceback, is  $O(n2^k)$ .*

*Proof.* A labeling is a partition of  $[k]$  into the subsets assigned to the modes, with empty subsets allowed. Its discrepancy is the maximum of their costs by Lemma 2.1. For the first  $i$  modes assigning precisely  $S$ , enumerate the subset  $U$  assigned to mode  $i$  and optimize the rest using the preceding modes. This proves the recurrence by induction. Each of  $2^k$  costs for one mode takes  $O(k)$  operations, and  $\sum_{S \subseteq [k]} 2^{|S|} = 3^k$  counts its transitions. Keep two objective rows and the minimizing subset for each state and mode. Backtracking from  $D_n([k])$  then gives a labeling with the stated storage.  $\square$

**Theorem 13.4** (Exact fixed-budget optimization). *Let  $s \geq 0$  and  $k = \min(s + 1, N)$ . An optimal grid schedule with at most  $s$  switches is computable in*

$$O\left(nN + \binom{N-1}{k-1} n(k2^k + 3^k)\right) \quad (114)$$

*arithmetic operations and  $O(nN + n2^k)$  storage. In particular, two switches admit  $O(nN^2)$  exact optimization. Repeated modes are permitted.*

*Proof.* Enumerate all  $\binom{N-1}{k-1}$  sets of internal block boundaries and use Proposition 13.3. Every resulting labeling has at most  $k - 1 \leq s$  changes. Conversely, any feasible grid schedule has at most  $k$  maximal runs. Subdivide these runs at unused grid boundaries until there are exactly  $k$  positive nominal blocks, assigning the same label to adjacent pieces. This is possible because  $k \leq N$  and does not change the control. Thus the enumeration covers every feasible schedule. Retain only the best answer and reuse the workspace for successive partitions.  $\square$

For fixed  $s$ , all rational numbers formed by this algorithm are sums or differences of polynomially many input numbers; comparisons do not enlarge them. Their bit lengths remain polynomial. The method is linear in  $n$  in its arithmetic count for fixed block count, but its factor of order  $N^{k-1}$  prevents a fixed-parameter tractability claim in  $s$  alone. Switching-time enumeration and exact graph methods for CIA are established [13, Section 6.4.3][3]. The subset recurrence is a standard partition dynamic program; its useful specialization here groups all occurrences of a mode, avoiding enumeration of the  $n^k$  possible mode words.

### 13.3 Dwell constraints and a candidate-label bound

Let each mode have a nonnegative minimum dwell duration  $d_i$ . We require every maximal constant run, including initial and terminal runs, to have length at least  $d_i$  when its mode is  $i$ . An unused mode has no dwell obligation. In (112), set  $c_i(U) = +\infty$  if a maximal string of consecutive indices in  $U$  has total block length below  $d_i$ . Consecutive assigned blocks must be merged before applying this test. The empty subset remains admissible. This check takes  $O(k)$  operations, so both complexity bounds remain valid. The subset recurrence is still exact because this admissibility depends only on  $i$  and  $U$ . Moreover, subdivision of a run preserves its physical length and hence its feasibility, so the proof of Theorem 13.4 remains valid. The algorithm detects infeasibility; for example, if every  $d_i > T$ , no schedule exists.

Other restrictions depending only on one mode's assigned subset can be treated by the same infinite-cost convention, with their checking cost included. Extending such a rule from prescribed blocks to the full-budget enumeration additionally requires invariance under subdivision into adjacent equally labeled blocks. Arbitrary transition graphs couple different modes and are outside this separation argument.

An alternative fixes both the block boundaries and their equality pattern. There are then  $d \leq k$  distinct roles, each comprising a nonempty subset of blocks, to be assigned injectively to modes. Let  $c_{ri}$  be the full-horizon component error when role  $r$  is assigned to mode  $i$ ; unselected modes have cost  $m_i$ .

**Lemma 13.5** (A bounded candidate set). *Suppose  $n \geq d$ . In the unrestricted role-assignment problem, some optimal assignment uses only the union of the  $d$  largest-mass modes and, for each role  $r$ , the  $d$  modes with smallest  $c_{ri}$ . This union has at most  $d^2 + d$  elements. Ties may be broken consistently in any way.*

*Proof.* Start with an optimal assignment using a label  $i$  outside that union. Since  $i$  is not among the  $d$  largest-mass labels and there are only  $d$  assigned roles, one of those  $d$  largest labels is omitted. Hence  $m_i$  is no larger than the current omitted-mass maximum. Among the  $d$  cheapest labels for  $i$ 's role, one label  $j$  is unused by the other  $d - 1$  roles. It differs from  $i$  and has no larger role cost. Replace  $i$  by  $j$ . The newly omitted mass  $m_i$  is bounded by the old omitted maximum, so the objective cannot increase. This removes one label outside the union and adds none. Iteration proves the claim. If  $n = d$ , all labels already belong to the union.  $\square$

This elementary exchange bound is useful for searches over equality patterns. The subset algorithm does not need it. No assertion of a new general assignment principle or unrestricted constraint support is intended.

### 13.4 Exact continuous optimization for piecewise-affine input

For completeness, a finite exact benchmark also exists when switches may occur inside input cells. Suppose  $A$  is piecewise affine on an  $N$ -cell rational input grid, and let  $k = s + 1$  without truncating it at  $N$ . Fix a word  $p \in [n]^k$  and a nondecreasing list of  $k - 1$  input-cell indices, one for each switch time. Write  $t_0 = 0, t_k = T$ . On its assigned cell,  $A_i(t_j)$  is affine in  $t_j$ . The service at that endpoint is

$$W_i(t_j) = \sum_{h=1}^j \mathbf{1}_{\{p_h=i\}}(t_h - t_{h-1}).$$

Therefore the cell bounds,  $t_1 \leq \dots \leq t_{k-1}$ ,  $0 \leq E \leq T$ , and

$$-E \leq A_i(t_j) - W_i(t_j) \leq E \quad (i \in [n], 1 \leq j \leq k) \quad (115)$$

form a rational linear program in  $k$  variables, minimizing  $E$ . Lemma 2.1 proves that no additional error constraints at input breakpoints are needed. For the one-sided objective retain only  $W_i(t_j) - A_i(t_j) \leq E$ .

There are  $n^k \binom{N+k-2}{k-1}$  word/cell choices. They cover all schedules: zero-length blocks and repeated labels are allowed, and a switch at an input boundary may be assigned to either incident closed cell. The minimum of their LP values is consequently the continuous instance optimum. Each feasible LP is bounded and has  $O(nk + k)$  inequalities. One entirely rational solution method enumerates all  $k$ -element subsets of its inequalities, solves the corresponding nonsingular equality systems, and retains the best feasible vertex. A nonempty bounded polytope has an optimal vertex with  $k$  independent active constraints, including when its affine dimension is smaller than  $k$ . Thus this method is complete. Its operation count adds a factor  $\binom{O(nk+k)}{k}$  and polynomial work per vertex, and its bit complexity is polynomial for fixed  $k$  by rational elimination and determinant bounds. This deliberately small-instance benchmark is less efficient in the number of modes than the grid subset algorithm. Its enumeration principle is classical; the explicit time-cell decomposition states how rational certification can be made complete.

## 14 Sharp grid transfer and certified coarsening

### 14.1 Supported rounding preserves the switch budget

Integral prefix rounding and support-preserving matching constructions are established tools. In particular, Besthorn and Kirches [2, Corollary 2.7] prove a uniform-mesh error bound with positive-support restrictions. We give the short flow construction needed here, then use temporal support to preserve the switch budget of a supplied integer schedule.

**Theorem 14.1** (Uniform-grid transfer). *Let  $W$  be a cumulative integer schedule with finitely many switches on  $[0, T]$ , and let  $\mathcal{T} = (0, h, \dots, Nh)$  with  $T = Nh$  and  $h > 0$ . There is a grid schedule  $V$  with no more switches than  $W$  such that*

$$\|V - W\|_\infty < h. \quad (116)$$

*Its word is obtained from a chronological subsequence of the original block word by merging adjacent equal labels. With rational original switch times and  $h$ , it can be constructed in polynomial bit complexity in the explicit grid length, number of modes, and input size.*

*Proof.* Set  $a_{ji} = h^{-1}(W_i(jh) - W_i((j-1)h))$  and  $S_{ji} = \sum_{q=1}^j a_{qi}$ . Each row of  $a$  is a simplex vector. Create a supply-one node for each cell and a chronological chain for each mode. The cell node may send flow to mode  $i$ 's chain at that cell only when  $a_{ji} > 0$ . The edge carrying mode  $i$ 's accumulated flow after cell  $j$  has integer lower and upper capacities  $\lfloor S_{ji} \rfloor$  and  $\lceil S_{ji} \rceil$ ; final chain edges enter a sink of demand  $N$ . The flows  $a_{ji}$  and  $S_{ji}$  show fractional feasibility. Integral capacitated-flow feasibility therefore gives assignments  $y_{ji} \in \{0, 1\}$  with exactly one per cell, supported where  $a_{ji} > 0$ , and

$$\lfloor S_{ji} \rfloor \leq \sum_{q=1}^j y_{qi} \leq \lceil S_{ji} \rceil.$$

Their distance from  $S_{ji}$  is strictly below one, including zero distance when  $S_{ji}$  is integral. Hence the endpoint errors are strictly below  $h$ . Within a cell, the selected component of  $V - W$  is nondecreasing and every other component is nonincreasing. The finitely many endpoint bounds prove (116) on the whole interval.

For each cell choose an interior time, away from the original switches, at which the original schedule occupies the selected mode. Positive cell occupation guarantees such a time. These times are strictly increasing, so their original block indices are nondecreasing. Remove repeated selections of the same original block. The selected word is then a subsequence of the original word; merging adjacent equal labels only decreases its length. It has no more changes than the original word. The network has polynomial size and integer capacities at most  $N$ ; standard integral flow algorithms and exact rational occupation calculations give the computational claim.  $\square$

**Corollary 14.2** (Instance and minimax transfer). *For every  $A \in \mathcal{A}_n(T)$ , switch budget  $s \geq 0$ , and uniform grid of width  $h$ ,*

$$\text{OPT}_s(A) \leq \text{OPT}_s^T(A) < \text{OPT}_s(A) + h. \quad (117)$$

*The minimax values, with the input classes in (6) and (7), satisfy*

$$F_{n,s}(T) \leq F_{n,s}^T < F_{n,s}(T) + h. \quad (118)$$

*Proof.* Restriction of the schedule set gives the first instance inequality. Apply Theorem 14.1 to an attained continuous optimum and use the triangle inequality for the strict upper bound. The first minimax inequality is (8), which uses exact cell averaging. For the other inequality, take a grid minimax maximizer  $A_*$ , whose existence follows from Proposition 2.2. Then  $F_{n,s}^T = \text{OPT}_s^T(A_*) < \text{OPT}_s(A_*) + h \leq F_{n,s}(T) + h$ . Attainment is essential to this argument for strictness; a supremum of arbitrary pointwise strict inequalities need not remain strict.  $\square$

The construction transfers a given schedule; it does not find its continuous optimum. It preserves neither arbitrary minimum dwell times nor arbitrary transition restrictions, since skipping a mode can create a forbidden transition. The multicomponent theorem uses a uniform grid: unequal cell lengths do not provide the unit assignment capacities used in the proof.

**Proposition 14.3** (Binary transfer on a nonuniform grid). *For two modes on any grid, a finitely switching schedule has a grid replacement with no more switches and discrepancy at most  $\Delta/2$ . This constant is sharp for transfer of a supplied schedule.*

*Proof.* Let  $D = \bar{\Delta}$  and maintain the mode-one endpoint discrepancy  $e = V_1 - W_1$  in  $[-D/2, D/2]$ . On a cell of length  $d \leq D$ , let  $a$  be the original mode-one occupation. If  $a = 0$  select mode two; if  $a = d$  select mode one. These supported choices leave  $e$  unchanged. Otherwise, both modes occur, and the two next discrepancies are  $e - a$  and  $e + d - a$ . The first is at most  $D/2$ , the second at least  $-D/2$ , and their distance is  $d \leq D$ . They cannot both lie outside the permitted interval, so choose one inside. The other component's discrepancy is its negative. Cellwise monotonicity and the support chronology in Theorem 14.1 finish the proof. On a single cell of length  $D$ , an original schedule occupying each mode for half the cell has discrepancy  $D/2$  from either possible constant grid schedule.  $\square$

**Proposition 14.4** (Sharp universal coefficient). *The coefficient one in (117) cannot be reduced uniformly over mode counts and budgets, even for grid-constant relaxed inputs and  $1 \leq s \leq N - 2$ .*

*Proof.* Use unit cells,  $n \geq 2$ ,  $N = n + 1$ , and  $s = n - 1 = N - 2$ . Let the relaxed control be uniform on  $[0, 1]$  and equal to mode  $n$  afterward. Every grid schedule incurs error  $1 - 1/n$  at time one in its first selected mode. The constant mode- $n$  schedule attains that error, so its grid optimum is  $1 - 1/n$ . A continuous schedule visits modes  $1, \dots, n$ , each for time  $1/n$  in the first cell, then remains in mode  $n$ . It has  $n - 1$  switches. All errors vanish at time one; within the first cell the largest absolute error is  $(n - 1)/n^2$ . Thus

$$\text{OPT}_s^T(A) - \text{OPT}_s(A) \geq (1 - 1/n) - (n - 1)/n^2 = (1 - 1/n)^2 \rightarrow 1.$$

Scaling proves the assertion for any mesh width  $h$ .  $\square$

Repeating the within-cell cycle  $m$  times, taking  $N = nm + 1$  and  $s = nm - 1$ , gives the additional family with continuous competitor error  $(n - 1)/(n^2 m)$  and gap at least  $(1 - 1/n)(1 - 1/(nm))$ . Already  $n = 4, m = 1$  gives a gap of at least  $9/16 > 1/2$ . This is an instance-wise counterexample to the half-mesh justification immediately preceding Conjecture 1 in Sager and Zeile [9, p. 615], also present in Zeile [13, p. 139]. Those passages motivate a conjecture; they are not stated as proved transfer theorems. The example does not identify the difference between two minimax values. For one switch, Corollary 3.5 retains the stronger nearest-boundary half-mesh estimate.

## 14.2 An additive certificate from exact coarse optimization

**Theorem 14.5** (Certified coarsening). *Let  $A \in \mathcal{A}_n(T)$  and fix a uniform coarse grid of  $M \geq 1$  cells,  $h = T/M$ . Compute its exact grid optimum  $U_h$  and an attaining schedule  $V_h$  using the exact cumulative values  $A_i(jh)$ . Then*

$$U_h - h < \text{OPT}_s(A) \leq U_h = D(A, V_h). \quad (119)$$

*In particular, for  $0 < \varepsilon \leq 1$  and  $M = \lceil 1/\varepsilon \rceil$ , this returns a feasible schedule whose suboptimality is strictly below  $\varepsilon T$ . If  $A$  is specified by rational piecewise-constant rates on an  $N$ -cell input grid, the coarse data can be formed in  $O(n(N + M))$  rational operations, even when the grids are not aligned. For each fixed  $s$ , the resulting additive approximation algorithm has polynomial bit complexity in the input length and  $1/\varepsilon$ .*

*Proof.* Endpoint monotonicity evaluates  $D(A, V_h)$  exactly from the coarse cumulative values; the rates need not be constant in coarse cells. Equation (117) proves the interval. Since  $h \leq \varepsilon T$ , it also proves the additive error claim.

For piecewise-constant input, traverse the sorted input and coarse endpoints together. On each segment of their common refinement, add its length times the known input rate

to the current coarse mass. The common refinement has at most  $N + M - 1$  cells, so this takes  $O(n(N + M))$  operations. This integration preserves the original  $A_i(jh)$ ; merely grouping whole fine cells would be incorrect for nonaligned grids. Apply Theorem 13.4 with  $k = \min(s + 1, M)$ , or the faster one-switch specialization when applicable. The arithmetic bound is

$$O\left(n(N + M) + \binom{M - 1}{k - 1} n(k2^k + 3^k)\right). \quad (120)$$

All integrations and subsequent computations use rationals of polynomial bit length. For fixed  $s$ , the bound is polynomial in  $N, n, M$  and input bit length. More generally its dependence on  $(s, \lceil 1/\varepsilon \rceil)$  can be placed in a parameter-only factor multiplying a polynomial in the input length.  $\square$

There are useful sharper special cases. For  $s = 1$ , moving an attained continuous optimum's single switch to a nearest grid endpoint changes its cumulative occupation by at most  $h/2$ . For  $n = 2$  and any  $s$ , Proposition 14.3 gives the same estimate. In either case,

$$U_h - h/2 \leq \text{OPT}_s(A) \leq U_h,$$

so  $M = \lceil 1/(2\varepsilon) \rceil$  suffices for additive error at most  $\varepsilon T$ . These lower inequalities are non-strict. With no switches, the grid and continuous instance optima agree exactly. The implementation uses the universal certificate (119) for a single uniform interface.

The lower endpoint may be replaced by zero when  $U_h - h < 0$ , but then the inequality at that endpoint is non-strict. If  $U_h - h = 0$ , strictness is retained. This is an additive scheme measured against  $T$ , not a relative approximation scheme: an instance optimum can be zero. If the coarse grid is a subset of a specified fine switching grid  $\mathcal{T}_f$ , the same certificate also brackets the fine optimum, since

$$U_h - h < \text{OPT}_s(A) \leq \text{OPT}_s^{\mathcal{T}_f}(A) \leq U_h.$$

For nonnested grids, the coarse schedule remains feasible for continuous switching, but need not be feasible for the given fine switching grid. The lower certificate uses switch-preserving transfer and thus does not extend automatically to dwell-constrained or transition-constrained optima.

**Example 14.6** (Dwell can prevent convergence under refinement). Take  $T = 1$ , two modes, one switch, and minimum dwell  $1/2$  for both modes. Let the input be pure mode one before  $1/2$  and pure mode two afterward. The identical continuous schedule is feasible and has error zero. On a uniform grid with an odd number  $M$  of cells, any feasible two-run schedule would require both runs to have length exactly  $1/2$ , which is not a grid boundary. Only constants are feasible, and their optimum error is  $1/2$ . Thus even along arbitrarily fine odd grids the dwell-constrained gap remains  $1/2$ . This gives a failure of an unrestricted transfer conclusion for these constraints, rather than merely a limitation of one construction.

**Corollary 14.7** (Input perturbation). *Suppose the data instead describe  $\tilde{A} \in \mathcal{A}_n(T)$  with a known bound  $\|A - \tilde{A}\|_\infty \leq \delta$ . Let  $U_h$  and  $V_h$  be the exact coarse optimum and schedule for  $\tilde{A}$ . Then*

$$U_h - h - \delta < \text{OPT}_s(A) \leq D(A, V_h) \leq U_h + \delta. \quad (121)$$

*Consequently the returned schedule has suboptimality strictly below  $h + 2\delta$ . If the pointwise rate error is bounded by  $\rho$ , one may use  $\delta = \rho T$ .*

*Proof.* Proposition 2.2 gives  $\text{OPT}_s(A) \geq \text{OPT}_s(\tilde{A}) - \delta$ ; combine it with the strict lower certificate for  $\tilde{A}$ . The schedule upper bound follows from the triangle inequality. Integrating a pointwise componentwise rate bound gives the stated cumulative bound.  $\square$

Exact optimization of quantized input is therefore distinguished from exact optimization of the original data. As above, clipping a negative lower endpoint to zero changes it to a non-strict endpoint. These certificates require no unknown continuous optimum and produce a feasible schedule along with explicit lower and upper bounds.

## 15 Exact reproducible computations

The computations check the formulas and constructions, compare independently implemented exact methods, and illustrate the cost of grid restriction. They use rational arithmetic throughout optimization and verification. Floating point is used only for plotting and elapsed-time reporting. The supplied verification bundle records source hashes, exact certificate data, input transformations, complete rational table values, and all timing samples. The computer-assisted theorems depend on the exact certificate checks described in Sections 6 and 12; the experiments below do not substitute for those universal proofs.

### 15.1 A pinned public relaxed-control profile

We use the three-mode Lotka–Volterra relaxed-control profile distributed with `pycombina` [8]: 12,000 cells of width  $10^{-3}$  on  $[0, 12]$ . The script retrieves a fixed commit, verifies its SHA-256 digest, and reads the published decimal entries as rationals. On each cell, it first normalizes the three nonnegative rates to sum to one, then applies largest remainder rounding with denominator  $10^6$ , breaking equal remainders by mode index. The maximum rate change in the normalization step is less than  $4.959 \cdot 10^{-7}$ . Quantization changes each normalized rate by less than  $10^{-6}$  and therefore changes cumulative allocations by less than

$$\delta = 12 \cdot 10^{-6}.$$

All exact optima reported here concern this quantized piecewise-constant input. Proposition 2.2 bounds the change of each optimum relative to the normalized decimal source by  $\delta$ . The unnormalized decimal columns are not taken as simplex-valued input.

The exact one-switch optimum on all 12,000 cells is 1.889, attained by switching from mode two to mode three at time 1.889. A separate enumeration of every ordered mode pair and every allowed boundary gives the same value. Direct evaluation at every original input endpoint verifies the returned schedule. The continuous one-switch optimum for the same input is

$$\text{OPT}_1(A) = \frac{4721469}{2500000} = 1.8885876, \quad \tau = 1.8885876, \quad (122)$$

with the same mode order. To certify this value, the script locates the unique crossing of  $t - A_p(t)$  and  $T - m_q - t$  for each of the six ordered pairs, using the exact affine formula in its input cell. Their maximum is minimized at that crossing; the omitted terminal mass is constant. The three-term formula therefore certifies all pair minima and their global minimum. Constants are included as endpoint choices. A second evaluation at all input knots and at the chosen switch verifies the attained error independently. The fine-grid gap is  $1031/2500000 = 0.0004124$ , below the one-switch half-mesh bound 0.0005.

Table 4 compares budgets on common coarser grids. Their boundaries are contained in the 12,000-cell grid. Coarse cell masses are exact sums of the fine masses, and every

Table 4: Exact quantized public-profile optima on specified uniform switching grids. All displayed decimals terminate and are exact. Every row uses one fixed grid for all four budgets; the original input is integrated exactly.

| Allowed cells $M$ | Width $h$ | $s = 0$   | $s = 1$   | $s = 2$   | $s = 3$    |
|-------------------|-----------|-----------|-----------|-----------|------------|
| 12                | 1         | 3.7771752 | 1.9999996 | 1.8540056 | 0.8540056  |
| 24                | 0.5       | 3.7771752 | 1.9999996 | 1.5       | 0.5263358  |
| 48                | 0.25      | 3.7771752 | 1.9999996 | 1.5       | 0.52633545 |

Table 5: Certified intervals for both the continuous optimum and the 12,000-cell optimum of the quantized public input. Each interval uses only its own coarse optimum and the general width- $h$  certificate. Parentheses indicate a strict lower endpoint.

| Allowed cells $M$ | Two switches           | Three switches           |
|-------------------|------------------------|--------------------------|
| 12                | (0.8540056, 1.8540056] | [0, 0.8540056]           |
| 24                | (1, 1.5]               | (0.0263358, 0.5263358]   |
| 48                | (1.25, 1.5]            | (0.27633545, 0.52633545] |

output schedule is expanded to the original grid and checked there. Independent labeled-word and boundary enumeration also verifies every entry for  $M = 12, 24$ . The  $M = 48$  entries are computed by the exact subset algorithm; its general correctness is established in Theorem 13.4. Increasing the budget on each fixed grid lowers the optimum. With three switches, refining from 12 to 24 cells gives a much larger improvement than refining from 24 to 48 cells. These are input-specific observations, not universal rates or comparisons with a solver using a different budget.

The intervals in Table 5 follow from Theorem 14.5. To certify the normalized decimal source instead, subtract  $\delta$  from each unclipped lower endpoint, add  $\delta$  to the upper endpoint, and then clip a negative lower bound to zero. The returned schedule’s error for that source is bounded by the enlarged upper endpoint. The one-switch rows admit the stronger non-strict lower bound  $U_h - h/2$ ; with no switches the grid optimum is already the continuous optimum. Thus these special cases do not need the more conservative general interval.

## 15.2 An exactly known continuous comparison

For the uniform three-mode input on  $[0, 1]$  with two switches, the continuous instance optimum equals  $1/6$ . Here is a direct proof, since Theorem 3.1 assumes  $k < n$ . Omitting a mode incurs error at least  $1/3$ . Otherwise all three modes occur once. If the last mode starts at  $v$ , its start and final errors imply

$$E \geq \max\{v/3, 2/3 - v\} \geq 1/6.$$

The word  $(1, 2, 3)$  with switches at  $1/4$  and  $1/2$  attains  $1/6$  by endpoint evaluation. Figure 2 compares this exact value with coarse optima for  $M = 3, 4, 5, 6, 8, 9, 12, 16, 24$  and their certified lower endpoints. Each finite-grid optimum is corroborated by separate labeled-word and boundary enumeration, and the returned schedule is checked directly. The coarse optima need not decrease between nonnested grids:  $M = 4$  attains  $1/6$ , whereas  $M = 5$  gives  $1/5$ . The certificate remains valid on every grid.

## 15.3 Controlled timings and implementation scope

Timings use Python 3.13.11 and exact `Fraction` arithmetic on an Intel Xeon w5-2565X, Linux under WSL2. Each reported number is the median of three calls in one process.

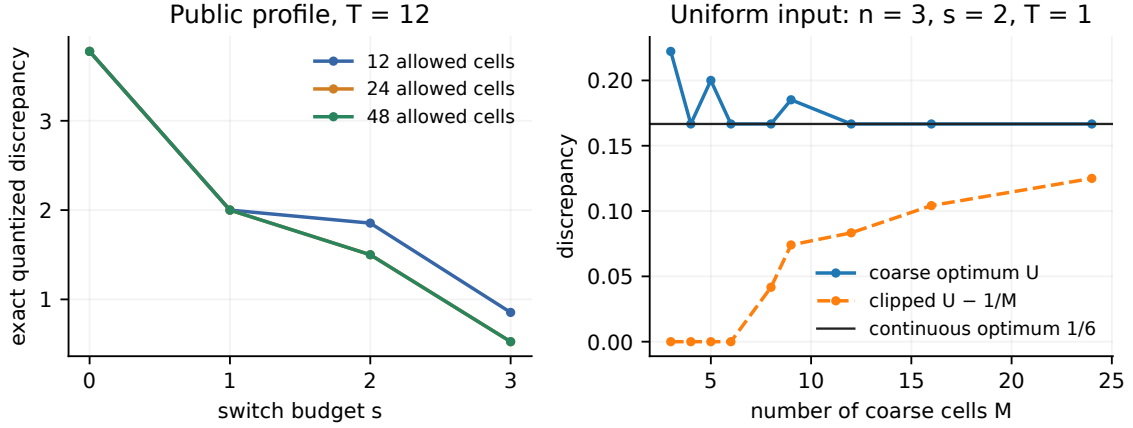


Figure 2: Left: exact optima on three nested switching grids for the same quantized public input. Right: coarse upper and clipped lower certificates for a uniform input with independently known continuous optimum. Lines join the displayed discrete cases only.

Table 6: Matched exact methods on deterministic rational inputs. Both methods optimize the same grid and switch budget. Times are medians in milliseconds; “words” counts all labeled-word/boundary cases, including repeated labels.

| $n$ | $N$ | $s$ | Words | Subset DP |       | Word enumeration |       |
|-----|-----|-----|-------|-----------|-------|------------------|-------|
|     |     |     |       | CPU       | Wall  | CPU              | Wall  |
| 3   | 8   | 2   | 567   | 3.85      | 3.82  | 7.48             | 7.42  |
| 4   | 8   | 2   | 1344  | 6.67      | 6.62  | 25.44            | 25.25 |
| 3   | 8   | 3   | 2835  | 14.02     | 13.92 | 47.71            | 47.36 |

Inputs are prepared before the timer starts; validation inside the solver is included, and independent verification afterward is excluded. Both CPU and wall times and all individual samples are archived. The host is shared, with no processor affinity or frequency control. These measurements describe this implementation and workload rather than a general performance ordering. The full 12,000-cell one-switch solve takes 0.246 seconds of CPU time and 0.249 seconds of wall time. The three-switch coarse solves take 0.085, 0.833, and 7.980 seconds of wall time for  $M = 12, 24, 48$ , respectively.

For Table 6, cell  $j = 0, \dots, N - 1$  has width  $1/N$  and rates proportional to  $1 + ((j + 2)(i + 3) + i^2 \bmod 11)$  for  $i = 0, \dots, n - 1$ . Both methods use exact rationals, and every optimum agrees. The word enumerator computes endpoint errors independently rather than calling the subset-cost routine. This is a controlled comparison with exhaustive enumeration, not with SCARP, `pycombina`, or the dwell-constrained heuristic in [1].

Table 7 illustrates the separate mode and grid costs at fixed budget. The  $O(nN^2)$  arithmetic bound for two switches is proved in Section 13; a small timing table cannot establish an asymptotic complexity claim. Rational bit lengths and interpreter overhead also affect these measurements. Higher budgets increase the boundary combinations substantially, which is the reason for using a coarse grid with an explicit accuracy certificate. The experiments do not solve the underlying nonlinear Lotka–Volterra optimization problem or measure its trajectory, objective, or path-constraint performance.

Table 7: Fixed-budget scaling cases: uniform inputs, horizon one, two switches. The first four rows vary modes at  $N = 12$ ; the remaining rows extend the  $n = 8$  grid-size comparison. Times are medians in milliseconds.

| Modes $n$ | Cells $N$ | CPU    | Wall   |
|-----------|-----------|--------|--------|
| 4         | 12        | 12.38  | 12.26  |
| 8         | 12        | 25.67  | 25.42  |
| 16        | 12        | 52.25  | 51.74  |
| 32        | 12        | 106.19 | 105.15 |
| 8         | 8         | 11.35  | 11.24  |
| 8         | 16        | 47.39  | 46.92  |
| 8         | 24        | 117.75 | 116.65 |

## 16 Scope and further questions

The continuous minimax formulas, finite-grid characterizations, and instance algorithms address different decisions. A minimax formula specifies the worst rounding error before the relaxed input is known. An exact instance algorithm finds the best schedule after the input is supplied. A coarsening certificate bounds the loss caused by reducing the allowed switching grid. Their combination separates switch-budget error, grid restriction, and input representation error without imposing smoothness on the original measurable control in the minimax analysis.

Three mathematical extensions remain substantial. First, the exact five-block one-sided reach bound for general mode count is unresolved. The exclusion identity in Appendix A identifies the missing weighted three-block inequality. The negative event assignment is not a control trajectory; chronological monotonicity removes that particular obstruction. A certificate for one chamber does not cover the remaining orderings or prove the full reach bound. The general and seeded results in Section 9 remain valid independently of this question. A successful extension would improve higher-budget minimax formulas and the second-order mode-count gap.

Second, the regime with as many or more activation blocks than modes has a different structure. The three-mode one-sided obstruction rules out a direct extension of the distinct-mode method. The exact full value  $F_{3,2}(T) = T/5$  and the range  $T/7 \leq F_{3,3}(T) \leq T/6$  show that finite-grid chamber methods can nevertheless improve the boundary theory. The floor-history construction is valid for every grid length, but the enumerations here stop at seven cells. The six exceptional seven-cell chambers have explicit repairs; their existence also disproves the tempting unit-error extension on  $2s + 1$  cells. Determining  $F_{3,3}$  and a general three-mode floor-history switch law requires further arguments. For other small-mode regimes, the classical bound (84) remains a useful comparison.

Third, constraints beyond a switch budget change the approximation question. The exact subset algorithm handles its stated per-mode minimum dwell rules, but a transition graph couples successive labels and is outside that unary assignment reduction. The odd-grid dwell example shows that unrestricted coarsening certificates can fail even under arbitrarily fine refinement. Useful constrained certificates would need assumptions ensuring feasible nearby switching times, or a constraint-aware refinement construction.

The algorithmic results already support exact small-budget rounding and certified additive approximation with rational data. A broader computational study could compare matched budgets and objectives against established graph and mixed integer methods, and then evaluate resulting trajectories under specified dynamics. Such a study would address application performance; it would not change the scope of the discrepancy theorems.

## A The remaining higher-block reach question

No result in the preceding sections assumes an exact reach theorem beyond four blocks. We now give the precise unresolved route and an exact test of one proposed relaxation. This separates a failed implication between linear inequalities from a counterexample generated by an actual control.

### A.1 An exclusion identity and its missing premise

Fix  $n \geq k + 1$ ,  $k \geq 3$ ,  $E > 0$ , and reach maps on  $[0, L]$ . For an available set  $U$  and  $h \leq |U|$ , let  $M_U^h$  be the maximum reach over all ordered  $h$ -tuples of distinct elements of  $U$ . Write  $M^h = M_{[n]}^h$ ,  $M_i^h = M_{[n] \setminus \{i\}}^h$ , and  $M_{ij}^h = M_{[n] \setminus \{i, j\}}^h$ . These are actual maxima of reach compositions, not independent event variables. Assume  $M^{k-1} < L$  so every event used below is uncapped.

**Lemma A.1** (General exclusion inequality). *Let  $S$  be a  $(k - 1)$ -element set supporting a word attaining  $M^{k-1}$ . Then*

$$(n - 2)M_i^{k-1} \geq nE + M_i^{k-2} + \sum_{j \neq i} M_{ij}^{k-2} - M^{k-1}, \quad (123)$$

$$\sum_i M_i^{k-1} \geq \left( n - k + 1 - \frac{k - 1}{n - 2} \right) M^{k-1} + \frac{(k - 1)nE + \mathcal{H}_S^{k-2}}{n - 2}, \quad (124)$$

$$\mathcal{H}_S^{k-2} := \sum_{i \in S} \left( M_i^{k-2} + \sum_{j \neq i} M_{ij}^{k-2} \right). \quad (125)$$

If also  $M^{k-1} \geq B_{k-1}$  and  $\mathcal{H}_S^{k-2} \geq (k - 1)nB_{k-2}$ , then  $k$  distinct blocks reach  $L$  whenever  $L \leq B_k$ .

*Proof.* Appending  $j$  to a maximizing  $(k - 2)$ -word avoiding  $i, j$  gives a reach no larger than  $M_i^{k-1}$ . By the uncapped boundary identity and monotonicity,

$$A_j(M_i^{k-1}) \leq M_i^{k-1} - E - M_{ij}^{k-2} \quad (j \neq i).$$

Sum over  $j \neq i$  and subtract from conservation to obtain

$$A_i(M_i^{k-1}) \geq (n - 1)E + \sum_{j \neq i} M_{ij}^{k-2} - (n - 2)M_i^{k-1}.$$

Appending  $i$  to a maximizing word counted by  $M_i^{k-2}$  likewise gives  $A_i(M^{k-1}) \leq M^{k-1} - E - M_i^{k-2}$ . Since  $M_i^{k-1} \leq M^{k-1}$ , monotonicity of  $A_i$  combines these bounds into (123). For  $i \notin S$  the same maximizing word remains available, so  $M_i^{k-1} = M^{k-1}$ . Sum (123) over  $i \in S$  and add these equal outside terms to obtain (124).

Its coefficient of  $M^{k-1}$  is positive for  $n \geq k + 1$ . Under the additional premises, use  $n(E + B_{k-2}) = (n - 1)B_{k-1}$  to deduce  $\sum_i M_i^{k-1} \geq nB_{k-1}$ . If no  $k$ -word reaches  $L$ , append  $i$  to a maximizing word avoiding  $i$ . Failure implies the strict inequality  $H_i(L) > M_i^{k-1} + E$ . Summing gives

$$(n - 1)L > \sum_i M_i^{k-1} + nE \geq nB_{k-1} + nE = (n - 1)B_k,$$

a contradiction to  $L \leq B_k$ . □

For  $k = 5$ , the premise  $M^4 \geq B_4$  is already established whenever  $M^4 < L$ . The missing premise in this route is the weighted three-block inequality  $\mathcal{H}_S^3 \geq 4nB_3$  for a maximizing four-mode set  $S$ . The stronger statement for every four-element  $S$  would suffice as well. For larger  $k$  both the corresponding weighted inequality and the preceding exact reach induction need justification. The research question is whether every  $n > k \geq 5$  measurable input admits  $k$  distinct blocks reaching  $\min\{T, B_k(n, E)\}$  for every  $E > 0$ . If true, it would identify  $G_{n,k}^-(T) = TL_{n,k}$  and, by the proved heavy-mode reduction, the full minimax. The question is unused, and the boundary example at  $n = k = 3$  does not answer it in the spare-mode range.

## A.2 A nonphysical feasible event assignment

Take  $n = 6$ ,  $E = 1$ , label modes  $0, \dots, 5$ , and fix  $S = \{0, 1, 2, 3\}$ . A natural relaxation stores six roots  $R_i$  and, for every available set  $U$  of size at least three, a pair event  $(2, U)$ ; for every  $|U| \geq 4$  it stores a triple event  $(3, U)$ . Each event  $v$  has time  $t_v$  and six allocations  $a_{v,i}$ . There are 42 pair events and 22 triple events, hence  $6 + 7 \cdot 64 = 454$  nonnegative variables. The full row specification is

$$\begin{aligned} R_i &\geq 1, \quad R_0 \leq R_i \ (i \neq 0), \quad \sum_{i \neq 0} R_i \geq 6, \\ \sum_i a_{v,i} &= t_v, \quad a_{(h,U),i} \geq R_i - 1 \quad (i \in U), \\ t_{(2,U)} - a_{(2,U),i} &\geq R_j + 1 \quad (i, j \in U, \ i \neq j), \\ t_{(3,U)} - a_{(3,U),i} &\geq t_{(2,U \setminus \{i\})} + 1 \quad (i \in U), \\ a_{(h,U),i} &\leq a_{(h',U'),i} \quad (h \leq h', \ U \subseteq U', \ (h, U) \neq (h', U')), \\ a_{(h,[6]),i} &\leq a_{(h,U),i} \quad (P_h \subseteq U \neq [6]), \quad P_2 = \{0, 1\}, \ P_3 = \{0, 1, 2\}. \end{aligned}$$

Here  $[6]$  denotes all six labels in this subsection. The last rows and the inclusion rows together force equality for events sharing the fixed global maximizing pair or triple. Each row is necessary for actual uncapped events with those maximizers and with  $R_0$  smallest. Indeed, adding available modes or activation blocks cannot decrease maximum reach, and coordinate allocations increase with time. In the all-available three-block objective define

$$\mathcal{H}_S^3 = \sum_{i \in S} \left( t_{(3,[6] \setminus \{i\})} + \sum_{j \neq i} t_{(3,[6] \setminus \{i,j\})} \right).$$

The bundled rational assignment satisfies all 3660 displayed inequalities and all 64 conservation equalities, with nonnegative variables, but gives

$$\mathcal{H}_S^3 = \frac{40328}{387} < \frac{13104}{125} = 4nB_3(6, 1). \quad (126)$$

Both a row-matrix checker and an independently organized event checker verify these facts exactly. Thus these necessary constraints do not imply the proposed weighted inequality for arbitrary four-element  $S$ .

This is not a counterexample to any reach theorem. The pair event for  $U = \{0, 2, 3\}$  has time  $1658/645$ ; that for  $V = \{2, 3, 4, 5\}$  has the later time  $614/215$ . Yet the corresponding mode 2 allocation falls from  $239/645$  to  $47/129$ . The time difference is  $184/645 > 0$  and the allocation drop is  $4/645 > 0$ . The sets are incomparable by inclusion, so the relaxation omitted this chronological comparison. A common cumulative control cannot produce these data. Nor does the relaxation require  $S$  itself to support a global maximizing four-word, a further condition if one only seeks the weaker premise in Lemma A.1.

**Lemma A.2** (Interpolation of event allocations). *Let finitely many events  $(t_v, a_v)$  have  $t_v \geq 0$ ,  $a_v \in \mathbb{R}_+^n$ , and  $\sum_i a_{v,i} = t_v$ . They lie on a common admissible cumulative trajectory if and only if*

$$t_v \leq t_w \implies a_{v,i} \leq a_{w,i} \text{ for every } i. \quad (127)$$

*Such a trajectory can be chosen piecewise affine, up to and beyond the last event.*

*Proof.* Necessity follows from coordinate monotonicity. For sufficiency, sort the events by time and add  $(0, 0)$ . Equal-time allocations agree because coordinate order in both directions forces equality; equivalently, nonnegative ordered differences with zero total are zero. Between distinct successive times interpolate linearly. Each coordinate increment is nonnegative and their sum is the time increment, so the slope vector is simplex-valued. Extend after the last event with any simplex-valued constant rate. This gives an admissible trajectory containing all event data.  $\square$

This lemma characterizes trajectory interpolation only. The variables  $R_i, t_{h,U}$  must additionally equal the specified latest inverse and maximum-composition reaches of that same trajectory. Conservation and chronological interpolation alone do not establish these identities.

### A.3 An exact chronological-chamber test

We strengthened the failed relaxation by sorting its 64 events in the nondecreasing time order of the saved witness. Ties are resolved by their original enumeration: first order  $h = 2, 3$ , then decreasing  $|U|$ , then lexicographic order of the increasing mode tuples. Let  $v_1, \dots, v_{64}$  be that fixed permutation. Add the 378 inequalities

$$a_{v_j,i} \leq a_{v_{j+1},i} \quad (1 \leq j < 64, 0 \leq i < 6). \quad (128)$$

Conservation then also forces  $t_{v_j} \leq t_{v_{j+1}}$ . Thus this is one closed chronological chamber, allowing ties, not a global imposition of the old witness's numerical times. The witness itself has 239 violations of ordered coordinate comparisons, with largest drop 224/645, and is excluded by these new rows.

**Proposition A.3** (One strengthened chamber, computer-assisted). *For this single fixed event permutation, the strengthened relaxation with 454 variables, 4038 inequalities, and 64 equalities has exact minimum*

$$\min \mathcal{H}_S^3 = 13104/125 = 4nB_3(6, 1).$$

*Proof.* Write the inequalities as  $Ax \leq b$ , the equalities as  $Qx = d$ , and the objective as  $c^\top x$ , with  $x \geq 0$ . The supplement supplies rational vectors  $y, z$  and verifies, component by component,

$$y \leq 0, \quad c - A^\top y - Q^\top z \geq 0, \quad b^\top y + d^\top z = 13104/125.$$

Consequently every feasible  $x$  satisfies  $c^\top x \geq y^\top Ax + z^\top Qx \geq b^\top y + d^\top z$ . There are 254 nonzero dual rows; the checker reconstructs every integer row from the bundled original relaxation and the explicit permutation, then sorts normalized sparse rows and their right-hand sides lexicographically before applying the dual vectors. It uses only rational arithmetic and does not invoke a numerical solver.

For attainment, take  $R_i = 6/5$ , every pair-event time  $66/25$ , every triple-event time  $546/125$ , and  $a_{v,i} = t_v/6$ . These are precisely the uniform-control events. In the chosen permutation all pair events precede all triple events, so every chamber row holds. All original rows and the objective equality are verified exactly as well. There are 24 triple-time terms in the objective, giving  $24(546/125) = 13104/125$ .  $\square$

The exact certificate closes the relaxation gap in this chamber. It does not cover the other chronological permutations, other choices of global maximizers, or other mode counts. It also does not claim that every point in the strengthened relaxation satisfies the inverse reach identities. A floating-point linear program was used only to discover the rational dual; the proof is the checked rational identity and the matching explicit primal. The general five-block reach problem remains unresolved by this investigation. Separately, the possible strengthening of Lemma 4.3 to force an adjacent pair of a largest heavy mode remains unused and unproved; it is independent of the established existential heavy-mode theorem.

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